Control Systems

Classroom Workbook

ELECTRONICS AND COMM. ENGINEERING

ELECTRICAL ENGINEERING

INSTRUMENTATION ENGINEERING

GATE/ESE

unacademy

Control Systems

Classroom Workbook

For ECE, EE and IN

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GATE Syllabus

ECE

Basic control system components; Feedback principle; Transfer function; Block diagram representation; Signal flow graph; Transient and steady-state analysis of LTI systems; Frequency response; Routh-Hurwitz and Nyquist stability criteria; Bode and root-locus plots; Lag, lead and lag-lead compensation; State variable model and solution of state equation of LTI systems.

EE

Mathematical modelling and representation of systems, Feedback principle, transfer function, Block diagrams and Signal flow graphs, Transient and Steadystate analysis of linear time invariant systems, Stability analysis using RouthHurwitz and Nyquist criteria, Bode plots, Root loci, Lag, Lead and Lead-Lag compensators; P, PI and PID controllers; State space model, Solution of state equations of LTI systems

IN

Feedback principles, signal flow graphs, transient response, steady-state-errors, Bode plot, phase and gain margins, Routh and Nyquist criteria, root loci, design of lead, lag and lead-lag compensators, state-space representation of systems; time-delay systems; mechanical, hydraulic and pneumatic system components, synchro pair, servo and stepper motors, servo valves; on-off, P, PI, PID, cascade, feedforward, and ratio controllers, tuning of PID controllers and sizing of control valves.

ESE Syllabus

E & T

Basic control system components; Feedback principle; Transfer function; Block diagram representation; Signal flow graph; Transient and steady-state analysis of LTI systems; Frequency response; Routh-Hurwitz and Nyquist stability criteria; Bode and root-locus plots; Lag, lead and lag-lead compensation; State variable model and solution of state equation of LTI systems.

EE

Transfer Function, Principal of Feedback, Block Diagram, Signal Flow Paragraphs, Steady-State Errors, Routh Hurwitz Creation, Nyquist Technology, Bode Plots, Root Loci, LED Leg Compensation, Stability Analysis, State Page Model, State Translation Matrix, Controllability and Observability, Linear State Variable Feedback, PID and Industrial Controller

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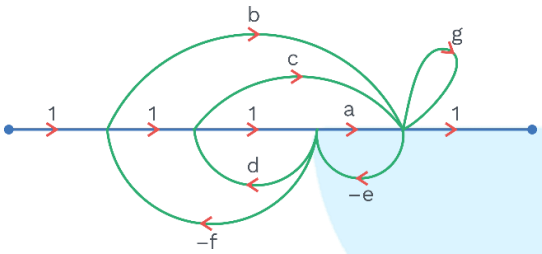
1. Mathematical Modeling, Block Diagram & SFG
2. Time Response Analysis
3. Routh Hurwitz Stability
4. Polar Plot and Root Locus Diagram
5. Nyquist Stability Criterion
6. Bode Plot
7. State Space Analysis
8. Controllers and Compensators

■■■■■

1 Mathematical Modeling, Block Diagram & SFG

MCQ, MSQ and NAT

Q.1 Determinant for the Signal Flow Graph shown below is



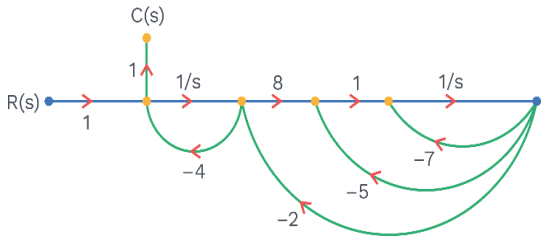
(A) $\Delta = 1 - d - g + f + ag - gf - ef - bef - cef$

(B) $\Delta = 1 + d - g - f - ae + gf + dg + bef - cef$

(C) $\Delta = 1 - d - g + f + ae - gf - dg + bef + ced - cef$

(D) $\Delta = 1 - d - g + f + ae - gf - dg - bef + ced - cef$

Q.2 The signal flow graph of a system is shown below. The transfer function $\frac{C(s)}{R(s)}$ of the system is-



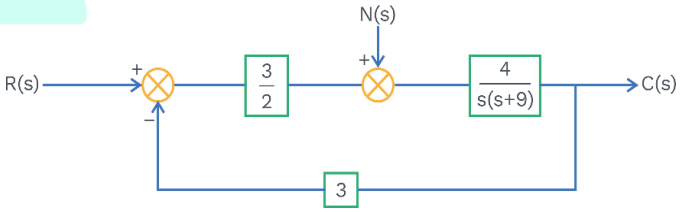
(A) $\frac{C(s)}{R(s)} = \frac{s(s+18)}{s^2+12s+28}$

(B) $\frac{C(s)}{R(s)} = \frac{(s+28)}{s^2+32s+48}$

(C) $\frac{C(s)}{R(s)} = \frac{s(s-18)}{s^2-12s+48}$

(D) $\frac{C(s)}{R(s)} = \frac{s(s+28)}{s^2+32s+48}$

Q.3 A block diagram is shown in the figure below:



Find the expression of c(t) if r(t) = e^{-2t} and n(t) = δ(t) is-

(A) $c(t) = \frac{3}{2}e^{-2t} - \frac{2}{3}e^{-3t} - \frac{5}{6}e^{-6t}$

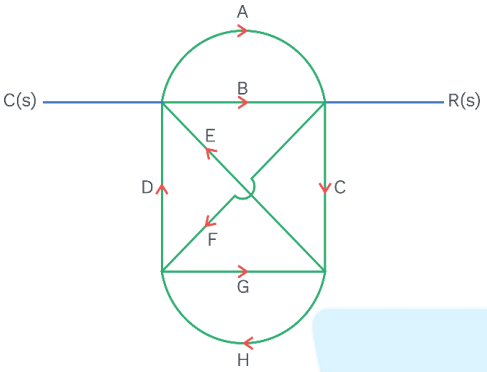
(B) $c(t) = \frac{5}{2}e^{-2t} - \frac{2}{3}e^{-3t} + \frac{7}{6}e^{-6t}$

(C) $c(t) = \frac{1}{2}e^{-9t} - \frac{3}{2}e^{-3t} - \frac{5}{2}e^{-4t}$

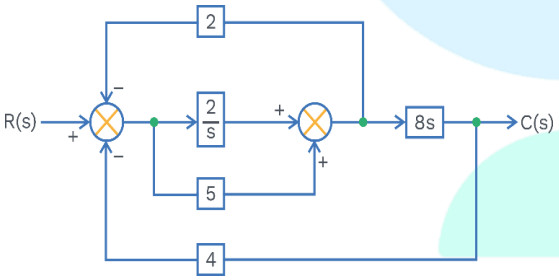
(D) $c(t) = \frac{1}{2}e^{-2t} + \frac{3}{2}e^{-3t} + \frac{5}{2}e^{-6t}$

Q.4 The total number of loops in the given signal flow graph is _____.

(Rounded off to the nearest integer)



Q.5 The transfer function of the given block diagram is-



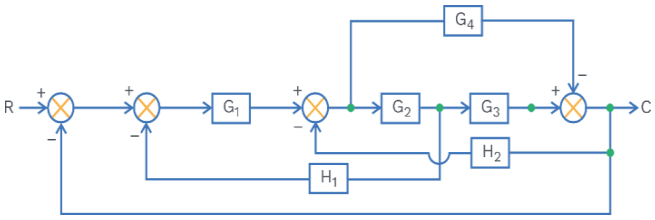
(A) $\frac{20s + 32}{60s^2 + 40s + 50}$

(B) $\frac{27s + 160}{20s^2 + 15s + 12}$

(C) $\frac{10s + 26}{240s^2 + 44s + 15}$

(D) $\frac{40s + 16}{160s^2 + 74s + 5}$

Q.6 Determine the $\frac{C}{R}$ for the given block diagram of a control system.



(A) $\frac{G_1G_2G_3 + G_1G_4}{1 + G_1G_2H_1 + G_2G_3H_2 + G_1G_2G_3 - G_1G_4 - G_4H_2}$

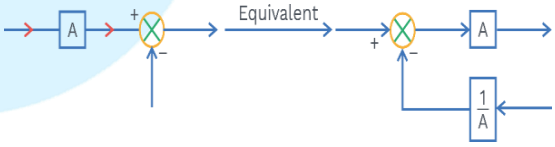
(B) $\frac{G_1G_2G_3 - G_1G_4}{1 + G_1G_2H_1 + G_2G_3H_2 + G_1G_2G_3 - G_1G_4 - G_4H_2}$

(C) $\frac{G_1G_2G_3 - G_1G_4}{1 + G_1G_2H_2 + G_2G_3H_1 + G_1G_2G_3 - G_1G_4 - G_4H_2}$

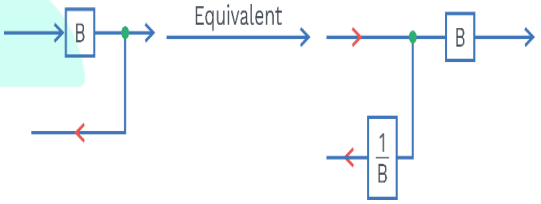
(D) $\frac{G_1G_2G_3 + G_1G_4}{1 + G_1G_2H_2 - G_2G_3H_1 - G_1G_2G_3 + G_1G_4 + G_4H_2}$

Q.7 Which of the rules and their equivalent block diagrams given in the options is/are correct for a control system?

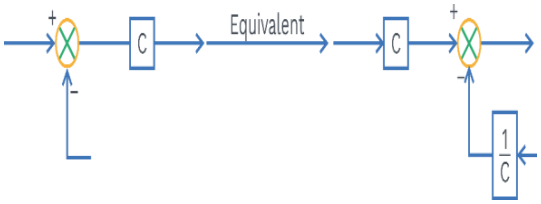
(A)



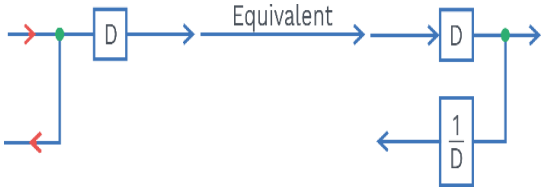
(B)



(C)



(D)



Q.8 A system is represented by the given equation, the transfer function of the system is-

$x = x_1 + B_3u$

$\dot{x}_1 = -A_1x_1 + x_2 + B_2u$

$\dot{x}_2 = -A_2x_1 + B_1u$

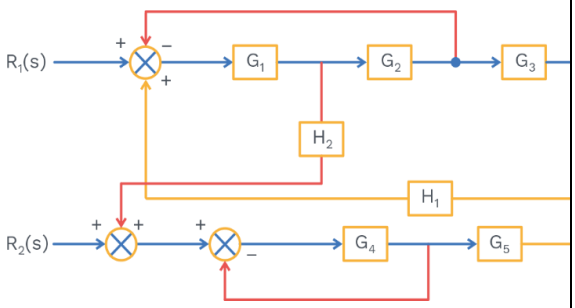
(A) $\frac{C(s)}{R(s)} = \frac{B_3 + B_2s + B_1(s^2 + A_1s + A_2)}{s^2 + A_1s + A_2}$

(B) $\frac{C(s)}{R(s)} = \frac{B_2 + B_1s + B_3(s^2 + A_2s + A_1)}{s^2 + A_1s + A_2}$

(C) $\frac{C(s)}{R(s)} = \frac{B_1 + B_2s + B_3(s^2 + A_2s + A_1)}{s^2 + A_2s + A_1}$

(D) $\frac{C(s)}{R(s)} = \frac{B_1 + B_2s + B_3(s^2 + A_1s + A_2)}{s^2 + A_1s + A_2}$

Q.9 The block diagram of the system given below has two inputs $R_1(s)$ and $R_2(s)$. If the input $R_1(s) = 0$, then the transfer function of the system is-



(A) $\frac{C(s)}{R_2(s)} = \frac{G_1G_2G_3 + G_4G_5}{1 + G_1G_2G_4 + G_1G_2 + G_1G_4H_1H_2}$

(B) $\frac{C(s)}{R_2(s)} = \frac{G_1G_2G_3G_4G_5H_1}{1 + G_4 + G_1G_2 + G_1G_2G_4 - G_1G_4G_5H_1H_2}$

(C) $\frac{C(s)}{R_2(s)} = \frac{G_1G_2G_3G_4G_5H_1}{1 - G_4 - G_1G_2 - G_1G_2G_4 + G_1G_4G_5H_1H_2}$

(D) $\frac{C(s)}{R_2(s)} = \frac{G_1G_2G_3 + G_4G_5}{1 - G_1G_2G_4 - G_1G_2 - G_1G_4H_1H_2}$

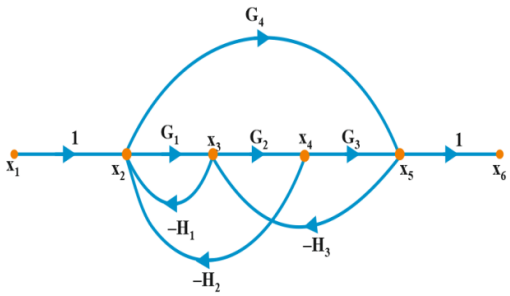
Q.10 For the given control system, the transfer function $\frac{C(s)}{R(s)}$ is-



(A) $\frac{1}{s+2}$ (B) $\frac{1}{s+1}$

(C) $\frac{s+1}{s^2+3s+2}$ (D) $\frac{1}{s^2+3s+3}$

Q.11 The number of loops available in the given signal flow graph is



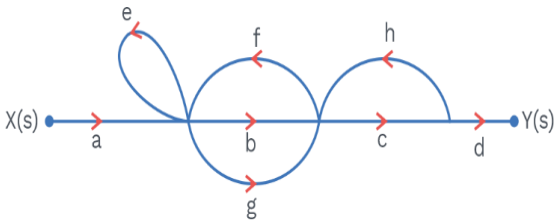
(A) 4

(B) 5

(C) 3

(D) 6

Q.12 The number of forward paths and the number of non-touching loop pairs in the given signal flow graph are respectively



(A) 2 and 3

(B) 2 and 1

(C) 1 and 2

(D) 3 and 2

Q.13 **Assertion (A) :** Sensitivity depends on the value of 1-GH.

Reason (R) : For negative feedback path, if the gain is positive then sensitivity decreases.

(A) Both (A) and (R) are true, and (R) is the correct explanation of (A).

(B) Both (A) and (R) are true, but (R) is not the correct explanation of (A).

(C) (A) is true, but (R) is false.

(D) (A) is false, but (R) is true.

Q.14 Transfer function of system described by

$$3\frac{d^2y(t)}{dt^2} + 4y(t) = \frac{dx(t)}{dt} + 2x(t)$$

(where x(t) is input signal and y(t) is output signal)

Consider the following statements.

(1) Transfer function = $\frac{3s^2 + 4}{s + 2}$

(2) Transfer function = $\frac{s + 2}{s^2 + 2}$

(3) DC gain = 0.5

(4) DC gain = 2

Which of the following is/are **incorrect**?

(A) 1 and 3 only

(B) 1 and 4 only

(C) 2, 3 and 4

(D) 1, 2 and 4 only

Q.15 If the closed-loop transfer function of a unity negative feedback system is $\frac{3}{s^2 + 9s + 17}$, find the corresponding open-loop transfer function.

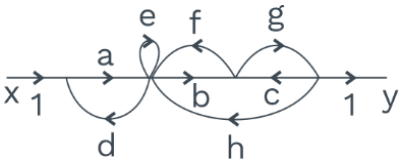
(A) $\frac{3}{s^2 + 9s + 14}$

(B) $\frac{3}{s^2 + 9s + 20}$

(C) $\frac{3}{s^2 + 9s}$

(D) $\frac{3}{s^2 + 9s + 17}$

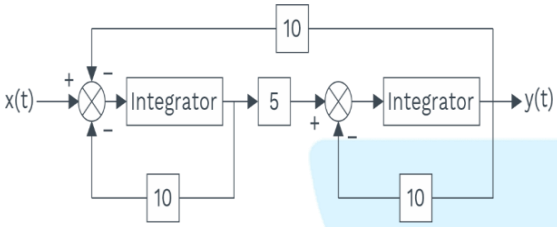
Q.16 In the figure below the number of forward paths and loops are



(A)1, 5 (B)1, 6

(C)2, 5 (D)2, 6

Q.17 The block diagram of a control system is shown is figure. The transfer function $G(s) = \frac{Y(s)}{X(s)}$ of the system is-



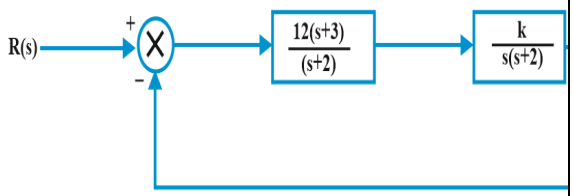
(A) $\frac{5}{s^2+20s+150}$

(B) $\frac{5}{s^2-20s+50}$

(C) $\frac{5}{s^2-20s-50}$

(D) $\frac{5}{s^2+20s+50}$

Q.18 Consider the feedback control system shown below –



Find the magnitude of the sensitivity of transfer function to variation in parameter K at $\omega = 6\,\text{rad/sec}$. (Assume process parameter K is 0.5).

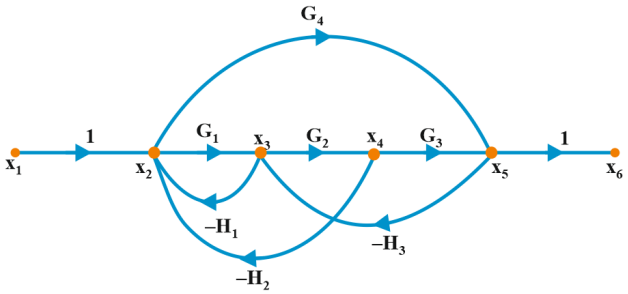
(A) 1.40

(B) 1.17

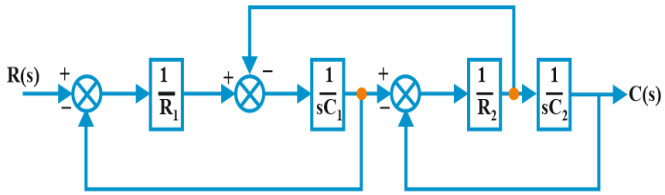
(C) 2.55

(D) 3.83

Q.19 The number of loops available in the given signal flow graph is _____. (Rounded off to nearest integer)



Q.20 Find the transfer function of the given block diagram –



(A)

$$\frac{C(s)}{R(s)} = \frac{1}{s^2R_1R_2C_1 + \left(R_1C_1 + R_2C_2 + R_1C_2\right)s + 1}$$

(B)

$$\frac{C(s)}{R(s)} = \frac{1}{s^2R_1R_2C_1C_2 + \left(R_1C_1 + R_2C_2 + R_1C_2\right)s + 1}$$

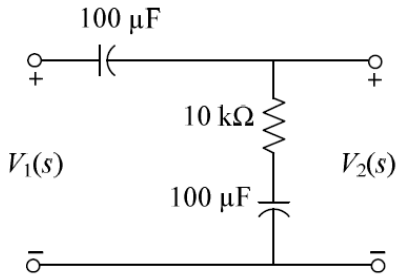
(C)

$$\frac{C(s)}{R(s)} = \frac{1}{s^2R_1R_2C_2 + R_1C_1 + R_2C_2 + R_1C_2s + 1}$$

(D)

$$\frac{C(s)}{R(s)} = \frac{1}{s^2R_1R_2C_1 + R_1C_1s + R_2C_2 + R_1C_2s + 1}$$

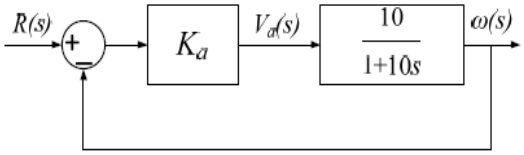
Q.21 The transfer function $\frac{V_2(s)}{V_1(s)}$ of the circuit shown below is



(A) $\frac{0.5s+1}{s+1}$

(C) $\frac{s+2}{s+1}$

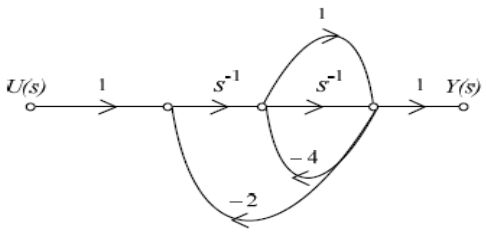
Q.22 The open-loop transfer function of a dc motor is given as $\frac{\omega(s)}{V_a(s)} = \frac{10}{1+10s}$. When connected in feedback as shown, the approximate value of K_a that will reduce the time constant of the closed loop system by one hundred times as compared to that of the open-loop system is



(A)1 (B)5

(C)10 (D)100

Q.23 The signal flow graph for a system is given below. The transfer function $\frac{Y(s)}{U(s)}$ for this system is



(A) $\frac{s+1}{5s^2+6s+2}$

(B) $\frac{s+1}{s^2+6s+2}$

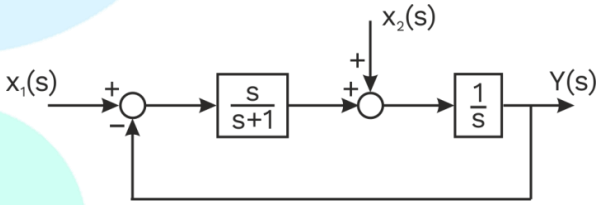
(C) $\frac{s+1}{3s+6}$

(B) $\frac{s^2+4s+2}{s+2}$

(D) $\frac{1}{s+2}$

(D) $\frac{s^2+1}{s+2}$

Q.24 For the following system

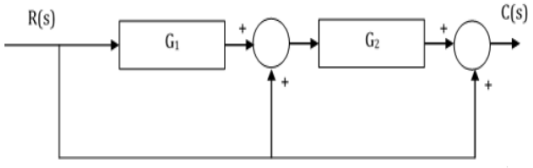


When $X_1(s) = 0$, the transfer function $\frac{Y(s)}{x_2(s)}$ is

(A) $\frac{s+1}{s^2}$ (B) $\frac{1}{s+1}$

(C) $\frac{s+2}{s(s+1)}$ (D) $\frac{s+1}{s(s+1)}$

Q.25 Consider the following block diagram in the figure.



The transfer function $\frac{C(s)}{R(s)}$ is

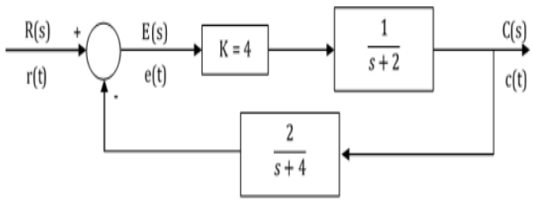
(A) $\frac{G_1G_2}{1+G_1G_2}$

(B) G_1 G_2+G_1+1

(C) $G_1G_2 + G_2 + 1$

(D) $\frac{G_1}{1+G_1G_2}$

Q.26 The steady state error of the system shown in the figure for a unit step input is _____.



Q.27 Let Y(s) be the unit-step response of a causal system having a transfer function,

$G(s) = \frac{3-s}{(s+1)(s+3)}$

That is, $Y(s) = \frac{G(s)}{s}$. The forced response of the system is:

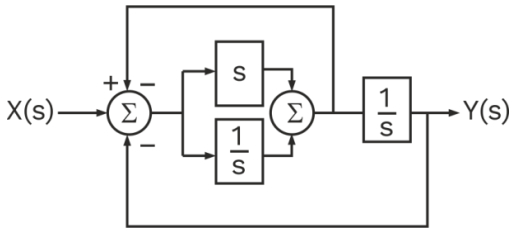
(A) $u(t) - 2e^{-t}u(t) + e^{-3t}u(t)$

(B) $2u(t)$

(C) $u(t)$

(D) $2u(t) - 2e^{-t}u(t) + e^{-3t}u(t)$

Q.28 The block diagram of a system is illustrated in the figure shown, where X(s) is the input and Y(s) is the output. The transfer function $H(s) = \frac{Y(s)}{X(s)}$ is:



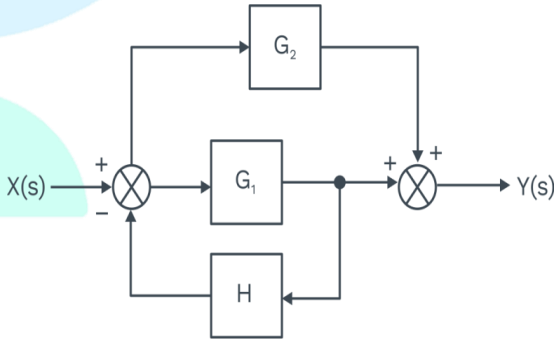
(A) $H(s) = \frac{s^2+1}{2s^2+1}$

(B) $H(s) = \frac{s^2+1}{s^3+2s^2+s+1}$

(C) $H(s) = \frac{s+1}{s^2+s+1}$

(D) $H(s) = \frac{s^2+1}{s^3+s^2+s+1}$

Q.29 The block diagram of a feedback control system is shown in the figure.



The transfer function $\frac{Y(s)}{X(s)}$ of the system is

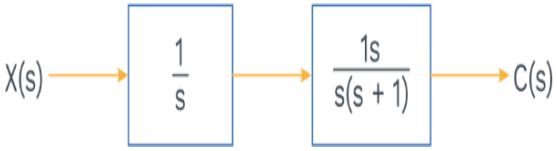
(A) $\frac{G_1+G_2+G_1G_2H}{1+G_1H}$

(B) $\frac{G_1+G_2}{1+G_1H+G_2H}$

(C) $\frac{G_1+G_2}{1+G_1H}$

(D) $\frac{G_1 + G_2 + G_1G_2H}{1 + G_1H + G_2H}$

Q.30 A system is represented by a block diagram as shown below –



The unit impulse response of the above system for time $t \geq 0$ is -

(A)$(1 - e^{t})\ u(t)$

(B)$(1 - e^{-t})\ u(t)$

(C)$(1 - e^{-t})\ u(t - 1)$

(D)$(1 + e^{-t})\ u(t)$

Q.31 A system is referred to as non-minimum phase system is -

(A)Poles and zeros do not lie on the right half of s-plane

(B)No zeros lie on the left half of s-plane

(C)No poles lie on the left half of s-plane

(D)Poles and zeros lie on the right half of s-plane

Q.32 A function is given as $F(s) = \frac{10}{s(s^2 + s + 4)}$, when F(s) is the Laplace transform of function f(t), the initial value of f(t) is -

(A)0 (B)10

(C)2.5 (D)0.4

Q.33 **Assertion (A):** In control system, the stability depends on the feedback.

Reason (R): If the system has positive feedback, the stability increases.

(A)Both A and R are true, and R is the correct explanation of A

(B)Both A and R are true, but R is not the correct explanation of A

(C)A is true, but R is false

(D)A is false, but R is true

Q.34 In a closed loop system negative feedback increases which of the parameter of the system.

(A)Overall Gain (B)Noise

(C)Bandwidth (D)Distortion

Q.35 Analogous of spring constant (K) in force current analogy-

(A)C

(B)1/C

(C)1

(D)1/L

Answer Key

Q.1 [D]

Q.2 [D]

Q.3 [A]

Q.4 [8]

Q.5 [D]

Q.6 [B]

Q.7 [A, D]

Q.8 [D]

Q.9 [B]

Q.10 [D]

Q.11 [B]

Q.12 [B]

Q.13 [D]

Q.14 [D]

Q.15 [A]

Q.16 [A]

Q.17 [A]

Q.18 [B]

Q.19 [5]

Q.20 [B]

Q.21 [D]

Q.22 [C]

Q.23 [A]

Q.24 [D]

Q.25 [C]

Q.26 [0.5, Range 0.49 to 0.51]

Q.27 [D]

Q.28 [B]

Q.29 [C]

Q.30 [B]

Q.31 [D]

Q.32 [A]

Q.33 [C]

Q.34 [C]

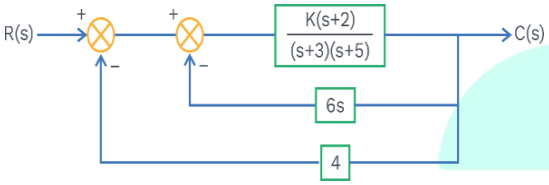
Q.35 [D]

2 Time Response Analysis

MCQ, MSQ and NAT

Q.1 A unity feedback system is having the closed-loop transfer function $\frac{C(s)}{R(s)} = \frac{6s+7}{(2s+3)\left(\frac{5}{2}s+6\right)}$. The steady-state error for the unit step input is _____. (Up to two decimal places)

Q.2 The block diagram of a control system is given below. The minimum value of K for critical damping is-



(A)K = 0.0268

(B)K = 0.6295

(C)K = –0.8567

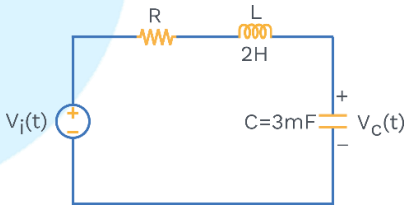
(D)K = –0.0182

Q.3 A unity feedback control system has an open-loop transfer function

$$OLTF = \frac{2s+5}{(s+6)(3s+2K)}$$

If the peak time for the second undershoot is $\frac{\pi}{6}$ second, then the value of |K| is _____. (Up to two decimal places)

Q.4 A circuit is shown below. The value of resistance (in ohms), if 10% of overshoot occurs in the voltage across the capacitor is_____ (Up to two decimal places)



Q.5 An open-loop transfer function of a unity feedback control system is given below:

$$G_s = \frac{K}{s(1+sT)}$$

If the damping ratio is increased from 0.3 to 0.9, then the amplifier gain changes from K_1 to K_2. Which of the following relation(s) is/are correct?

(A)If $K_2 = \frac{1}{2}$, then $K_1 = 4.5$

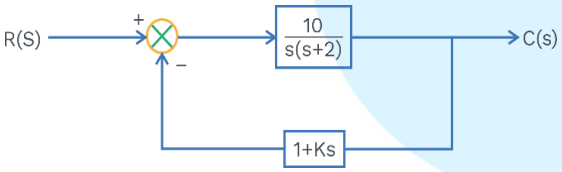
(B)If $K_2 = 9$, then $K_1 = 81$

(C) If $K_1 = 9$, then $K_2 = 4.5$

(D)If $K_1 = 45$, then $K_2 = 5$

Q.6 The open-loop transfer function of a unity feedback control system is given by $G(s) = \frac{58}{s^2(s+6)(2s^2+3s+15)}$. The steady-state error for input r(t) = $5+6t+3t^2$ is _____. (Up to two decimal places)

Q.7 The block diagram of a control system is given below. If the damping ratio is 0.5, then the value of K and rise time are respectively-



(A) K = 0.42 and t_r = 1.323 sec

(B) K = 0.525 and t_r = 1.33 sec

(C) K = 1.5 and t_r = 0.925 sec

(D K = 0.116 and t_r = 0.765 sec

Q.8 An open-loop transfer function of a system is given below as-

$$G(s) = \frac{16}{s^2+4s+16} \quad ; \quad H(s) = Ks$$

If the damping ratio of the system is 0.8, then -

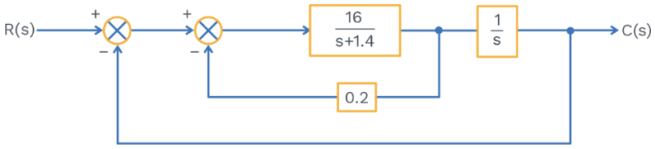
(A) Overshoot of the system is 1.52 %.

(B) K= 2.12

(C) Overshoot of the system is 3.22 %.

(D) K= 0.15

Q.9 The block diagram of a system is given below.



The rise time of the system is _____ sec. (Up to three decimal places)

Q.10 The impulse response of an initially relaxed linear system is $6e^{-3t}u(t)$. To produce a response of $te^{-3t}u(t)$,the input required is-

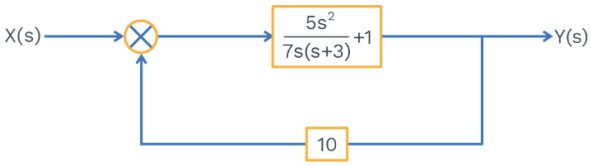
(A) $\frac{6e^{-3t}u(t)}{5}$

(B) $\frac{e^{-3t}u(t)}{6}$

(C) $\frac{e^{-3t}u(t)}{3}$

(D) $6e^{3t}u(t)$

Q.11 Calculate the DC gain of the closed-loop negative feedback system shown below.



(A) $\frac{2}{81}$

(B) 0

(C) $\frac{3}{127}$

(D) $\frac{1}{11}$

Q.12 The output of a control system is given as-

$c(t) = 1 + 0.5e^{-25t} - 1.5e^{-10t}$

then the damping ratio of the system for unit step input is _____ . (Up to three decimal places)

Q.13 The differential equation of a closed-loop transfer function is given by-

$$\frac{d^2c}{dt^2} + \frac{8dc}{dt} = 64(r - c)$$

The peak overshoot of the system is _____ %. (Up to one decimal place)

Q.14 The open loop transfer function of a plant is given as-

$$G(s) = \frac{50}{(s+0.2)(s+5)(s+50)}$$

The approximate open loop transfer function is obtained by the retention of one pole closest to the frequency response of the given transfer function at low frequency. The approximate OLTF is-

(A) $\frac{1}{s}$

(B) $\frac{0.2}{(s+0.2)}$

(C) $\frac{50}{(s+0.2)}$

(D) $\frac{1}{(s+0.2)(s+5)}$

Q.15 A system with transfer function

$$G(s) = \frac{(s^2+9)(s+2)}{(s+1)(s+3)(s+4)}$$

is excited by $\sin(\omega t)$ .The steady-state output of the system is zero at

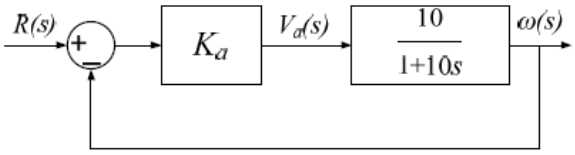
(A) $\omega = 1\ rad/s$

(B) $\omega = 2\ rad/s$

(C) $\omega = 3\ rad/s$

(D) $\omega = 4\ rad/s$

Q.16 The open-loop transfer function of a dc motor is given as $\frac{\omega(s)}{V_a(s)} = \frac{10}{1+10s}$. When connected in feedback as shown, the approximate value of K_a that will reduce the time constant of the closed loop system by one hundred times as compared to that of the open-loop system is



(A) 1 (B) 5

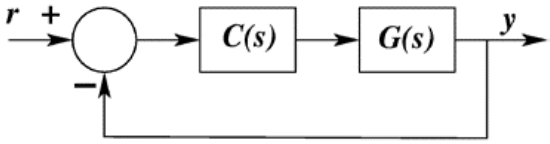
(C) 10 (D) 10

Q.17 The forward path transfer function of a unity negative feedback system is given by

$$G(s) = \frac{K}{(s+2)(s-1)}$$

The value of Kwhich will place both the poles of the closed-loop system at the same location, is _____.

Q.18 For the following feedback system $G(s) = \frac{1}{(s+1)(s+2)}$. The 2% settling time of the step response is required to be less than 2 seconds.



Which one of the following compensators C(s) achieves this?

(A) $3\left(\frac{1}{s+5}\right)$

(B) $5\left(\frac{0.03}{s}+1\right)$

(C)2(s + 4)

(D) $4\left(\frac{s+8}{s+3}\right)$

Q.19 The natural frequency of an undamped second-order system is 40 rad/s. If the system is damped with a damping ratio 0.3, the damped natural frequency in rad/s is _____

Q.20 The input $-3e^{2t}u(t)$, where (t) is the unit step function, is applied to a system with transfer function $\frac{s-2}{s+3}$. If the initial value of the output is −2, then the value of the output at steady state is _____.

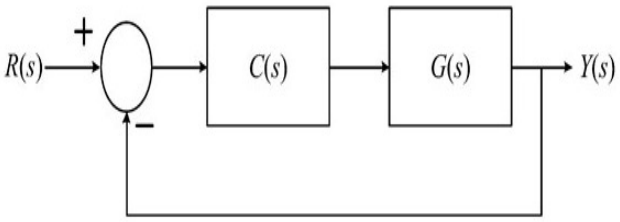
Q.21 Consider a causal second-order system with the transfer function $G(s) = \frac{1}{1+2s+s^2}$ with a unit-step $R(s) = \frac{1}{s}$ as an input. Let C(s) be the corresponding output. The time taken by the system output c(t) to reach 94% of its steady-state value $\lim_{t\to\infty} c(t)$, rounded off to two decimal places, is:

(A)5.25 (B)2.81

(C)4.50 (D)3.89

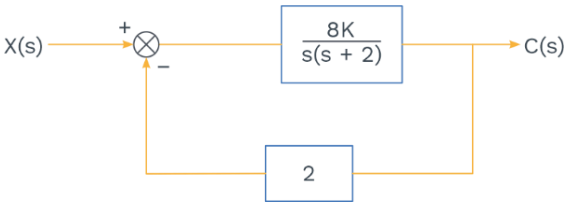
Q.22 A system with transfer function $G(s) = \frac{1}{(s+1)(s+a)}$, a>0 is subjected to an input 5 cos3t. The steady state output of the system is $\frac{1}{\sqrt{10}}\cos(3t-1.892)$. The value of a is _____

Q.23 Consider the following closed loop control system.



Where $G(s) = \frac{1}{s(s+1)}$ and C(s) = $K\frac{s+1}{s+3}$. If the steady state error for a unit ramp input is 0.1, then the value of K is _____.

Q.24 A control system has a closed loop transfer function as –



To make the response of the system non-oscillatory at constant frequency, the value of K is -

(A) 1.12 (B) 0.62

(C) 0.16 (D) 0.06

Q.25 Match List-I (Damping ratio) with List-II (Nature of system)

List-I	List-II
(P) $\xi < 1$	(i) Overdamped
(Q) $\xi > 1$	(ii) Critically damped
(R) $\xi = 1$	(iii) Underdamped
(S) $\xi = 0$	(iv) Undamped

Select the correct option using the code given below.

(A) (P) – (ii), (Q) – (i), (R) – (iv), (S) – (iii)

(B) (P) – (iv), (Q) – (i), (R) – (iii), (S) – (ii)

(C) (P) – (iii), (Q) – (i), (R) – (ii), (S) – (iv)

(D) (P) – (iv), (Q) – (i), (R) – (ii), (S) – (iii)

Q.26 In a unity negative feedback, the open loop transfer function is –

$$G(s) = \frac{K(s+1)}{s(1+s)(1+2s)}$$

The steady state error for unit ramp input when K = 6 is -

(A) 0.16 (B) 1.6

(C) 6 (D) 2

Q.27 The impulse response of an initially relaxed linear system is $3e^{-5t}u(t)$. To produce a response of $te^{-5t}u(t)$, the input should be-

(A) $\frac{0.2e^{-5t}u(t)}{3}$

(B) $\frac{e^{-5t}u(t)}{3}$

(C) $\frac{2e^{-5t}u(t)}{3}$

(D) $3e^{5t}u(t)$

Q.28 An LTI system is given as $2y''(t) + 12y'(t) + 5y(t) = 6x'(t) + x(t)$, the steady state value of the unit step response is-

(A) 3 (B) 0.2

(C) 5 (D) 1/3

Q.29 A closed loop control system pole and zero is located at –1–3j and –1+3j respectively. The undamped resonant frequency is-

(A) $\sqrt{80}$ rad/sec

(B) $\sqrt{10}$ rad/sec

(C) $2\sqrt{2}$ rad/sec

(D) $\sqrt{2}$ rad/sec

Q.30 A system has velocity error constant $K_v = 200$, then the system is-

(1) Type = 0 and steady state error $= \frac{1}{201}$ for unit step input.

(2) Type =1 and steady state error $= \frac{1}{201}$ for ramp input.

(3) Type = 1 and steady state error = 0 for unit step input.

(4) Type = 1 and steady state error $= \frac{1}{200}$ for ramp input.

Which of the option is correct-

(A) 1 and 2

(B) 2 and 3

(C) 3 and 4

(D) 1 and 4

Q.31 If we decrease the damping factor (ξ) of the system, then peak overshoot

1. Increases
2. Decreases
3. Remain constant
4. Makes the system more stable

Which of the statement(s) is/are correct?

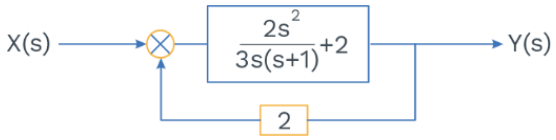
(A) Only 1

(B) Only 2

(C) Only 3

(D) 2 and 4

Q.32 Calculate the DC Gain of the closed loop negative feedback system shown below-



(A) 7/8

(B) 1/2

(C) 8/7

(D) 2

Answer Key

Q.1 [0.61, Range 0.58 to 0.63]

Q.2 [C]

Q.3 [25.90, Range 25 to 26.5]

Q.4 [30.52, Range 30.45 to 30.60]

Q.5 [A, B, D]

Q.6 [9.31, Range 9.28 to 9.35]

Q.7 [D]

Q.8 [A, D]

Q.9 [0.667, Range 0.63 to 0.69]

Q.10 [B]

Q.11 [D]

Q.12 [1.106, Range 0.98 to 1.12]

Q.13 [16.3, Range 16 to 16.5]

Q.14 [B]

Q.15 [C]

Q.16 [C]

Q.17 [2.25, Range 2.24 to 2.26]

Q.18 [C]

Q.19 [38.157, Range 37.5 to 39]

Q.20 [0, Range -0.01 to 0.01]

Q.21 [C]

Q.22 [4, Range 3.95 to 4.05]

Q.23 [30, Range 29.5 to 30.5]

Q.24 [D]

Q.25 [C]

Q.26 [A]

Q.27 [B]

Q.28 [B]

Q.29 [C]

Q.30 [C]

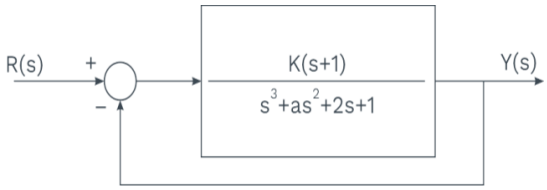
Q.31 [A]

Q.32 [D]

3 Routh-Hurwitz Stability

MCQ, MSQ and NAT

Q.1 The feedback system shown below oscillates at 2 rad/s when



(A) $K = 2$ and $a = 0.75$

(B) $K = 3$ and $a = 0.75$

(C) $K = 4$ and $a = 0.5$

(D) $K = 2$ and $a = 0.5$

Q.2 A polynomial

$$f(x) = a_4x^4 + a_3x^3 + a_2x^2 + a_1x - a_0$$

with all coefficients positive has

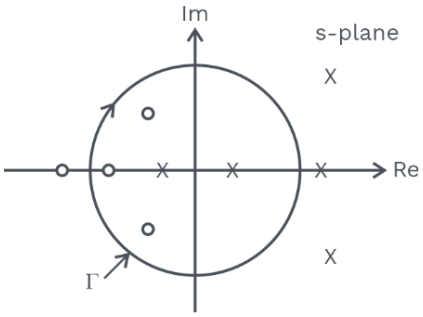
(A) no real roots

(B) no negative real root

(C) odd number of real roots

(D) at least one positive and one negative real root

Q.3 The pole-zero map of a rational function G(s) is shown below. When the closed contour Γ is mapped into the G(s) -plane, then the mapping encircles:



(A) the origin of the G(s) -plane once in the counter-clockwise direction.

(B) the origin of the G(s) -plane once in the clockwise direction.

(C) the point $-1+j0$ of the G(s) -plane once in the counter-clockwise direction

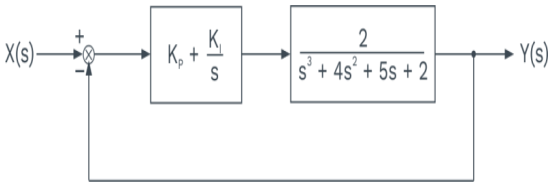
(D) the point $-1+j0$ of the G(s) -plane once in the clockwise direction.

Q.4 The loop transfer function of a negative feedback system is

$$G(s)\,H(s) = \frac{K(s+11)}{s(s+2)(s+8)}.$$

The value of K, for which the system is marginally stable, is _____.

Q.5 A unity feedback system that uses proportional-integral (PI) control is shown in the figure.



The stability of the overall system is controlled by tuning the PI control parameters K_P and K_I. The maximum value of K_I that can be chosen so as to keep the overall system stable or, in the worst case, marginally stable (rounded off to three decimal places) is _____.

Q.6 For a unity feedback control system, the open loop transfer function is given by $G(s) = \frac{e^{-2s}}{s(s+p)}$. If the system is stable, then the possible value of p is

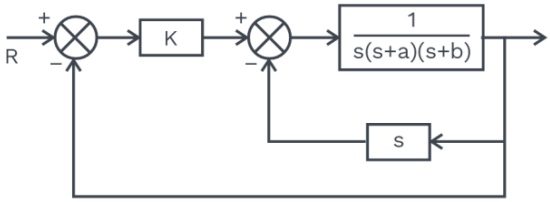
(A) 1.3

(B) 2.01

(C) 2

(D) 1

Q.7 Consider the unity feedback system which employs feedback as shown in the figure.



The frequency of oscillation of the above system is

(A) $\sqrt{ab} + 1$ rad/sec

(B) $\sqrt{ab}$ rad/sec

(C) $\sqrt{ab+1}$ rad/sec

(D) $ab + 1$ rad/sec

Q.8 Consider a unity negative feedback system, with open loop transfer function as $G(s) = \frac{K(s^2+1)}{(s+1)(s+a)}$ (a is positive constant). If the system is stable for K >–1, then which of the following is correct?

(A) a < 2 (B) 0 < a < 1

(C) a > 1 (D) No such value

Q.9 The number of roots with the positive real parts does the characteristic equation has –

$s^3 + s^2 + 2s + 1 = 0$

(A) One (B)Two

(C) Three (D)Zero

Q.10 The first column of Routh table consists of the terms –2, –4, 6, –2.2, 1. Find the number of unstable poles for the respective system.

(A) 1 (B) 2

(C) 3 (D) 4

Q.11 The characteristics equation of a system is given by $s^3 + s^2 + 9s + 9 = 0$. The undamped natural frequency of the system is _____ rad/sec. (Rounded off to nearest integer)

Q.12 The characteristic equation of the system is given as:

$q(s) = s^4 + 6s^3 + 14s^2 + 16s + 8$.

The largest time constant of the characteristic equation that is greater than 1 sec is _____sec. (Rounded off to nearest integer)

Q.13 The number of poles lying on the left-hand side of the jω axis for the characteristic equation $s^6 + 2s^5 + s^4 - s^2 - 2s - 1 = 0$ is _____. (Rounded off to nearest integer)

Q.14 The forward path transfer function for a unity negative feedback system is:

$$G(s) = \frac{Ke^{-0.2s}}{1+s}$$

The value of K for the system to be stable is-

(A) $K > 5$ (B) $-1 < K < 5$

(C) $K < -1$ (D) $-1 < K < 0$

Q.15 Forward path transfer function of a closed loop system is given as-

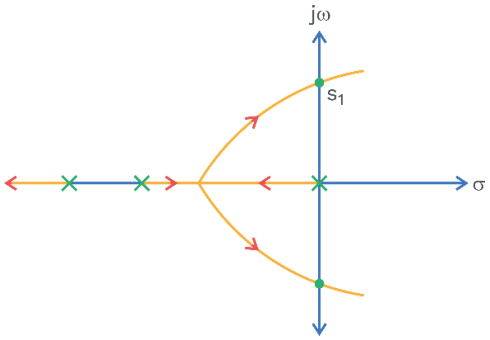
$$G(s) = \frac{K}{s(s+1)(s+9)} \; ; \; H(s) = 1$$

When the gain of the system is set at the finite value, the oscillation of the system occurs at _____ rad/sec. (Rounded off to nearest integer)

Q.16 The Routh array of a characteristic equation is given as-

s^3	1	y
s^2	x	z
s^1	$\frac{10-K}{5}$	
s^0	K	

The plot of the system is-



The coefficients x, y, z and K are greater than zero. For critical stability, the value of $\left|\frac{s_1}{K}\right|$ should be _____. (Up to three decimal place)

Q. 17 For the equations $s^3 - 4s^2 + 5 + 6 = 0$ the number of roots in the left half of s-plane will be

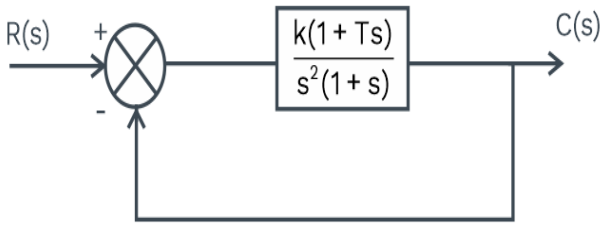
(A)Zero

(B)One

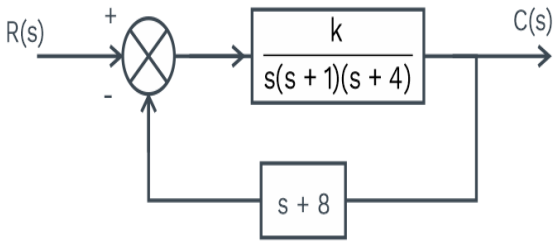
(C)Two

(D) Three

Q.18 For a feedback control system as shown in figures. The system is stable for all positive values of k (i.e. for $k > 0$), if the value of T is greater than _____



Q.19 Consider the system shown in the figure below. The value of system gain k for which of the system becomes oscillatory is _____.



Q.20 A unity feedback system having open loop gain, becomes stable when

$$G(s)H(s) = \frac{k(1-s)}{1+s}$$

(A) $|k| > 1$

(B) $k > 1$

(C) $|k| < 1$

(D) $k < -1$

Q.21 If the loop gain k of a negative feedback system having a transfer function $\frac{k(s+3)}{(s+8)^2}$, to be adjusted in induce a sustained oscillation then.

(A)The frequency of this oscillation must be $4\sqrt{1}$ rad/sec

(B)The frequency of this oscillation must be 4 rad/sec

(C)The frequency of this oscillation must be 4 (or) $4\sqrt{3}$ rad/sec

(D)Such a k does not exist

Q.22 Consider the following equation given below,

$s^3 + 3s^2 + 4s + A = 0.$

All the roots of this equation are in the left half of the s-plane provided that condition is___.

(A)$5 < A < 12$

(B) $A > 12$

(C) $-3 < A < 4$

(D)$0 < A < 12$

Q.23 For a control system, the characteristic equation is

$s^4 + 2s^3 + 3s^2 + 4s + 5 = 0$

Then the nature of the system and the number of poles at RHS respectively is …… .

(A) unstable, 3 (B) stable, 0

(C) unstable, 2 (D) unstable, 1

Q.24 A control system with the characteristic equation given as,

$s^4 + 3s^3 + 16s^2 + 6s + K = 0$

Then the value of K for which the system is unstable is …… .

(A) 14 (B) 7

(C) 30 (D) 26

Q.25 For what positive value of k does the polynomial,

$2s^4 + 8s^3 + 12s^2 + 24s + K = 0$

have roots with zero real parts?

Q.26 From the following statements given below, which of these are true regarding the Routh-Hurwitz criterion?

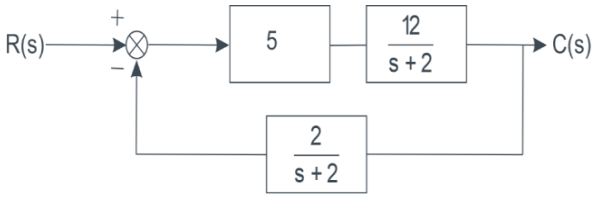
(A) It gives the number of closed loop poles lying on the right half of the s-plane.

(B) It gives number of zeros lying on the right half of s-plane.

(C) It gives the gain margin and the phase margin.

(D) It provide absolute stability.

Q.27 For the closed-loop system shown below,



Then the characteristic equation is -

(A) $s^2 + 2s + 120 = 0$

(B) $s^2 + 2s + 124 = 0$

(C) $s^2 + 4s + 124 = 0$

(D) $s^2 + s + 124 = 0$

Q.28 For a unity feedback control system, the open-loop transfer function is given as –

$$G(s) = \frac{K}{s(s+1)(s+2)},$$

Then the system is stable if,

(A) $0 < K < 6$

(B) $K > 6$

(C) $K > 0$

(D) system is always unstable

Q.29 A system with the characteristic equation is given by –

$s^6 + s^5 + 2s^4 + 3s^3 + 10s^2 + 4s + 2 = 0$

Then the number of the roots of the equation that lies on the imaginary axis of s-plane is -

(A) 4 (B) 3

(C) 0 (D) 2

Q.30 In Routh's tabular method, the first column elements are given as –

$$4, -2, 3, \frac{1}{4}, \frac{2}{7}$$

It means that

(A) There are two roots in the right half of s-plane.

(B) There is one root in the left half of s-plane.

(C) There is one root in right half of the s-plane.

(D) there are three roots in the left half of the s-plane.

Q.31 For the characteristic equation given below,

$s^4 + Ks^3 + s^2 + s + 1 = 0$

Then the range of k for stability is -

(A) $K > 1$ (B) $K < 1$

(C) $K > 0$ (D) none of these

Q.32 For a control system, the unity feedback system is given as –

$$G(s) = \frac{K}{(s+1)(s^2+s+1)}$$

Then the value of frequency of oscillation (in rad/sec) for a marginally stable system is____.

(A) $\frac{1}{\sqrt{2}}$ (B) 2

(C) $2\sqrt{2}$ (D) $\sqrt{2}$

Q.33 An LTI system having a characteristic equation in negative unity feedback configuration is shown below –

$s^4 + 2s^2 + 4 = 0$

Then the number of poles located on $j\omega$ axis is _____.

Q.34 The characteristic equation of a control system is $s^3 + 4s^2 + 6s + 13 = 0$

Then the number of roots of the equation which lie to the left of the line $s + 1 = 0$.

Q.35 The characteristics equation of a system is given by

$3s^4 + 10s^3 + 6s^2 + 11s + 2 = 0$

The system is

(A) stable

(B) marginal stable

(C) unstable

(D) insufficient data

Q.36 The value of 'K' for which unity feedback control system crosses the imaginary axis is_____.

Q.37 The number of roots of the characteristic equation that lies on the right half of the s-plane.

$3s^4 + 2s^3 + 3s^2 + 5s + 10 = 0$

Q.38 The stable unity feedback control system whose open loop transfer function is given below. The relation between K_1 and K_2 is

$$G(s) = \frac{K_1}{s(s(s+5)+K_2)}$$

(A) $K_1 < 10\ K_2$

(B) $0 < K_1 < 5K_2$

(C) $K_1 < 5K_2$

(D) $0 < K_1 < K_2$

Q.39 Which of the following characteristics polynomial of a system shows stable operation in the feedback system?

(A) $s^3 + 4s^2 - 6s + 1 = 0$

(B) $s^4 + 2s^3 + s + 9 = 0$

(C) $s^3 + 6s^2 + 5s + 30 = 0$

(D) $s^4 + 8s^3 + 24s^2 + 32s + 30 = 0$

Answer Key

Q.1 [A]

Q.2 [D]

Q.3 [B]

Q.4 [160, Range 159 to 161]

Q.5 [3.125, Range 3 to 3.2]

Q.6 [B]

Q.7 [C]

Q.8 [C]

Q.9 [B]

Q.10 [C]

Q.11 [3]

Q.12 [1]

Q.13 [3]

Q.14 [B]

Q.15 [3]

Q.16 [0.141, Range 0.1 to 0.2]

Q.17 [B]

Q.18 [1]

Q.19 [6.67, Range 6.5 to 6.75]

Q.20 [C]

Q.21 [B]

Q.22 [D]

Q.23 [C]

Q.24 [C]

Q.25 [18, Range 17.5 to 18.5]

Q.26 [A, D]

Q.27 [C]

Q.28 [A]

Q.29 [C]

Q.30 [A]

Q.31 [D]

Q.32 [D]

Q.33 [0]

Q.34 [1]

Q.35 [A]

Q.36 [120, Range 119 to 121}

Q.37 [2]

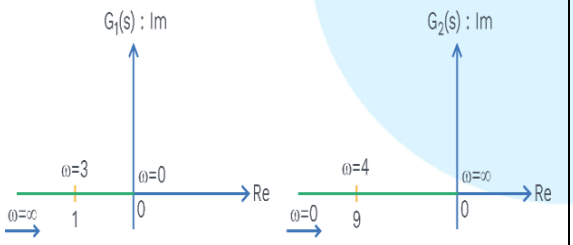
Q.38 [B]

Q.39 [D]

4 Polar Plot and Root Locus Diagram

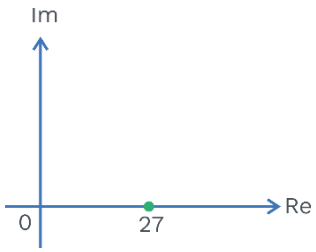
MCQ, MSQ and NAT

Q.1 The polar plots of two systems, $G_1(s)$ and $G_2(s)$ are shown below.

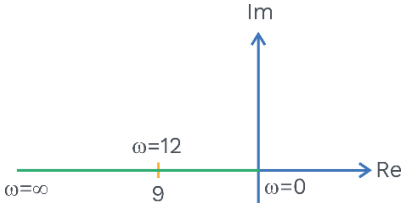


The result of $G_1(s)\,xG_2(s)$ is-

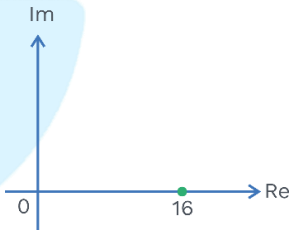
(A)



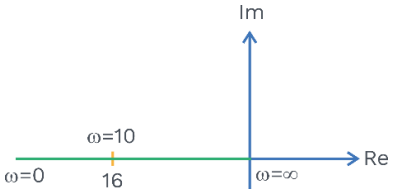
(B)



(c)



(D)

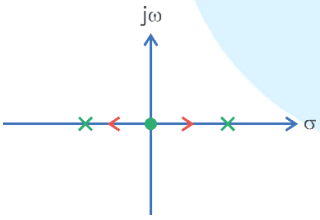


Q.2 A control system with unity feedback has a forward path transfer function $G(s)=\dfrac{K}{s(s+1)}$. If the closed-loop frequency response has a peak resonant magnitude of 4, then the 3dB bandwidth is _____rad/sec. (Up to three decimal places)

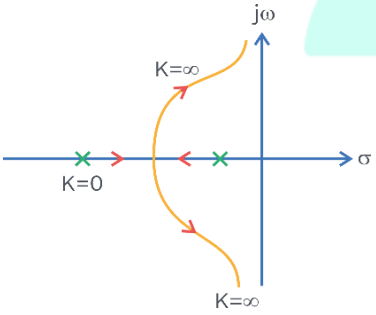
Q.3 The open loop transfer function for a system is given as $\frac{(3+7s+5s^2)}{(s+2)(s+1)(s^2+5s+7)}$. The point where the asymptotes intersect the real axis lies at _____. (Up to two decimal places)

Q.4 Determine the root locus of the system if closed loop transfer function is $T(s)=\frac{K}{s^2-(K-8)s+1}$.

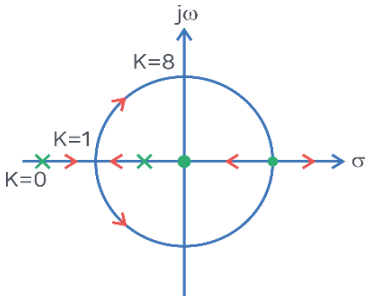
(A)



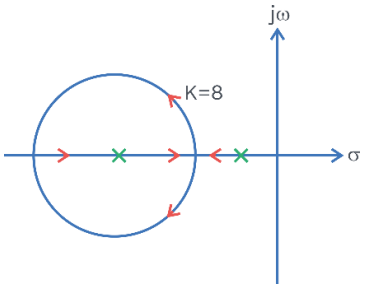
(B)



(C)



(D)

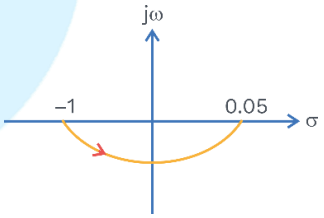


Q.5 The open-loop transfer of a system is-

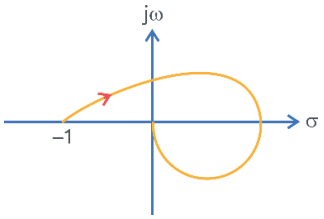
$$G(s)H(s)=\frac{(s+1)(1+0.2s)}{(s-1)(1+4s)}$$

The polar plot for the transfer function is-

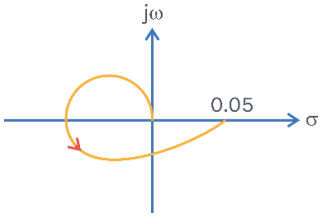
(A)



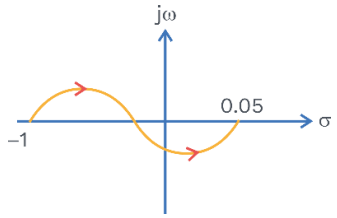
(B)



(C)



(D)



Q.6 The open-loop transfer function of a unity feedback control system is-

$$G(s) = \frac{K}{(s+6)(s+5)}$$

The value of K at the breakaway point of the root locus plot of the system is _____. (Up to two decimal places)

Q.7 An open loop system is given below. The centroid of the asymptotes is-

$$G(s) = \frac{k(s+2)}{s(s^2+2s+2)(s^2+2s+5)}$$

(A) –0.5 (B) –1

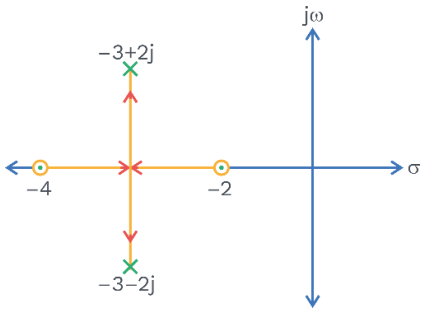
(C) 1 (D) 2

Q.8 The open loop transfer function of a system is given by-

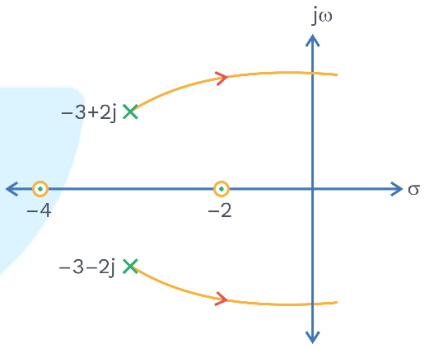
$$G(s)H(s) = \frac{K(s+2)(s+4)}{s^2+6s+13}$$

The root locus diagram is-

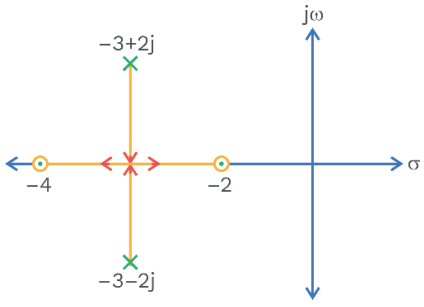
(A)



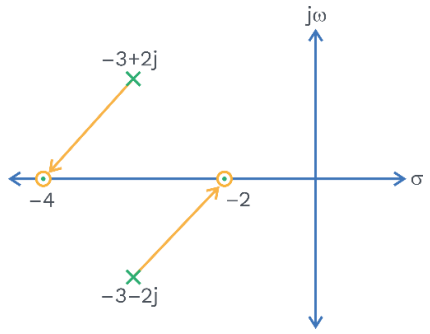
(B)



(C)

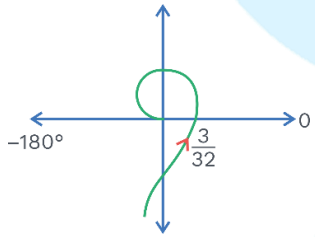


(D)

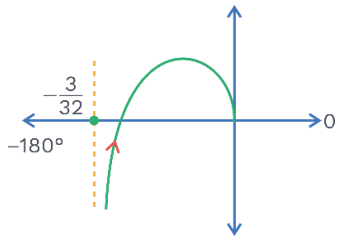


Q.9 The unity feedback function is given as $G(s)=\frac{1}{s(s+2)(s+4)}$. The plot of G(jω)H(jω) for 0 <ω<∞ is-

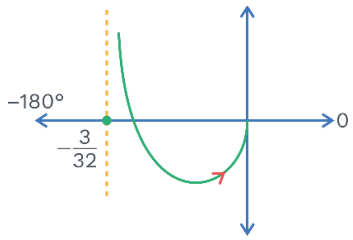
(A)



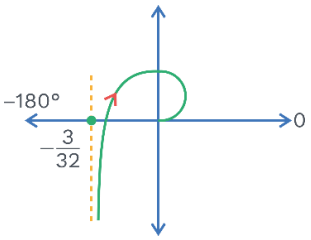
(B)



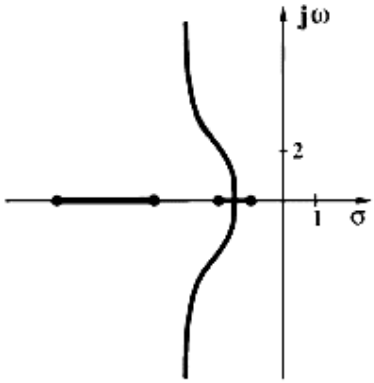
(C)



(D)



Q.10 In the root locus plot shown in the figure, the pole/zero marks and the arrows have been removed. Which one of the following transfer functions has this root locus?



(A) $\frac{s+1}{(s+2)(s+4)(s+7)}$

(B) $\frac{s+4}{(s+1)(s+2)(s+7)}$

(C) $\frac{s+7}{(s+1)(s+2)(s+4)}$

(D) $\frac{(s+1)(s+2)}{(s+7)(s+4)}$

Q.11 The characteristic equation of a system is

$s^3+3s^2+(K+2)s+3K=0$.

In the root locus plot for the given system, as K varies from 0 to □, the break-away or break-in point(s) lie within

(A) (–1, 0) (B) (–2, –1)

(C) (–3, –2) (D) (–∞, –3)

Q.12 Consider the following systems:

System 1: $G(s) = \frac{1}{(2s+1)}$

System 2: $G(s) = \frac{1}{(5s+1)}$

The true statement regarding the two given systems is

(A) Bandwidth of system 1 is greater than the bandwidth of system 2

(B) Bandwidth of system 1 is lower than the bandwidth of system 2

(C) Bandwidth of both the systems is same

(D) Bandwidth of both the systems is infinite

Q.13 Consider the following statements about Root locus:

S1 : The root locus is symmetrical with respect to both (real and imaginary) axis.

S2 : The root locus start from (K = 0) from the open loop poles/zeros and terminates (K = ∞) on either finite open loop zeros or infinity.

S3 : If root locus intersect at imaginary axis, the point of intersection are conjugate,

S4 : For higher values of K, root locus can be approximated by asymptotic lines and these asymptotic lines intersect at a point on Real axis.

Which of the above is correct?

(A) S1, S2, S3 and S4

(B) S3 and S4 only

(C) S2, S3 and S4 only

(D) S1 and S2 only

Q.14 The closed loop transfer function of a system is –

$$G(s)H(s) = \frac{K(s+8)}{(s+2)(s+3)}$$

Then the centroid of the root locus is located as -

(A) 4 (B) 3

(C) –3 (D) 2

Q.15 For the determination of the transient response and the stability of a system, the strongest method used is -

(A) Routh Hurwitz

(B) Polar plot

(C) Bode

(D) Root locus

Q.16 An open loop transfer function is given as-

$$G(s)H(s) = \frac{K(s+1)(s-5)}{(s-8)(s+3)(s-2)}$$

Find the centroid of the root locus.

(A) +3 (B) –2

(C) –1 (D) +2

Q.17 The open loop transfer function of a unity feedback system is-

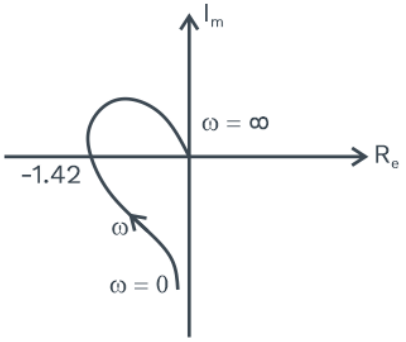
$$G(s)H(s) = \frac{K^2(2s+3)}{s^2(s+5)}$$

In root locus plot if K varies from 0 to ∞, find the value of K, when all roots are real and equal-

(A) $K = \frac{5\sqrt{5}}{9}$ (B) $K = \frac{125}{81}$

(C) $K = \frac{125}{8}$ (D) $K = \frac{3\sqrt{5}}{81}$

Q.18 The polar plot of an open loop stable system is shown below. The closed loop system is.



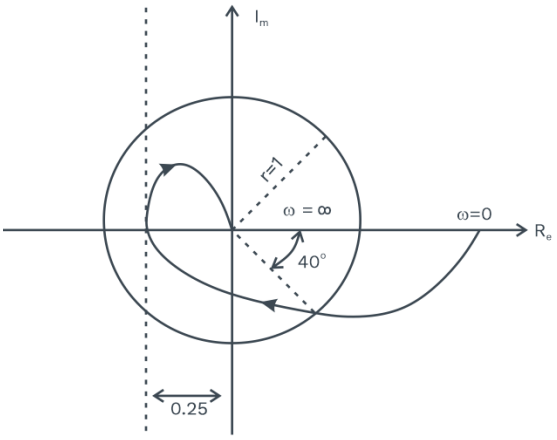
(A) Always stable

(B) Marginally stable

(C) Unstable with one pole on the RHS plane

(D) Unstable with two poles on the RHS plane

Q.19 The polar plot of a negative feedback, open loop control system as shown below



The gain margin & the phase margin of the system are respectively.

(A) 4, 60°

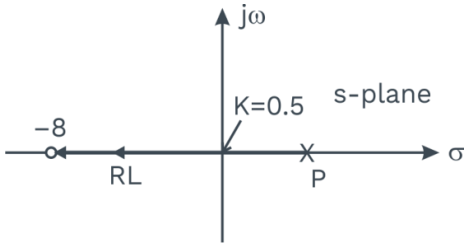
(B) 4, 140°

(C) 2, 60°

(D) 2, 140°

Q.20 The unity negative feedback control system has an open loop transfer function is $G(s) = \frac{k\left(s + \frac{4}{3}\right)}{s^2(s+12)}$, the values of 'k' such that all the closed loop poles are equal will be_____

Q.21 The root locus of a unity feedback control system is shown below,



Then the open loop transfer function is-

(A) $\frac{K(s+5)}{(s-2)}$ (B) $\frac{K(s+8)}{(s+4)}$

(C) $\frac{K(s-8)}{(s+4)}$ (D) $\frac{K(s+8)}{(s-4)}$

Q.22 Which of the following properties of the breakaway point in the root locus of a closed loop control system with characteristic equation $1 + KG(s)\,H(s) = 0$ is/are correct?

(A) At breakaway point, $\frac{dK}{ds} = 0$

(B) At breakaway point, $\frac{dK}{ds} \neq 0$

(C) The breakaway point can lie only on the real axis.

(D) At the breakaway point, K is maximum for the root locus on the real-axis.

Answer Key

Q.1 [C]

Q.2 [6.096, Range 6.0 to 6.180]

Q.3 [-3.3, Range -3.5 to -3.01]

Q.4 [C]

Q.5 [D]

Q.6 [60.5, Range 59.5 to 61.5]

Q.7 [A]

Q.8 [C]

Q.9 [B]

Q.10 [B]

Q.11 [A]

Q.12 [A]

Q.13 [B]

Q.14 [B]

Q.15 [D]

Q.16 [A]

Q.17 [A]

Q.18 [D]

Q.19 [B]

Q.20 [48, Range 47.5 to 48.5]

Q.21 [D]

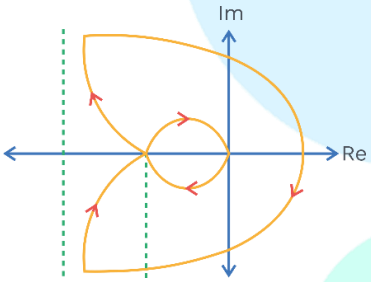
Q.22 [A, D]

5 Nyquist Stability Criterion

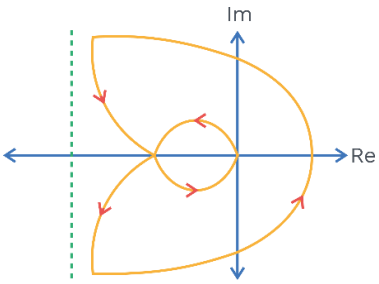
MCQ, MSQ and NAT

Q.1 The forward path transfer function is given as $G(s)=\frac{K}{s}$ that has feedback of $H(s)=\frac{1}{s^2+s+2}$.

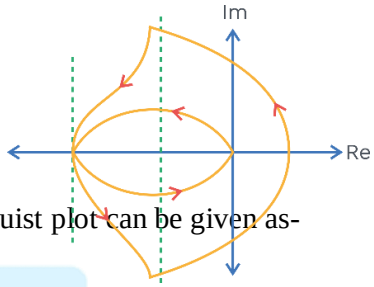
(A)



(B)

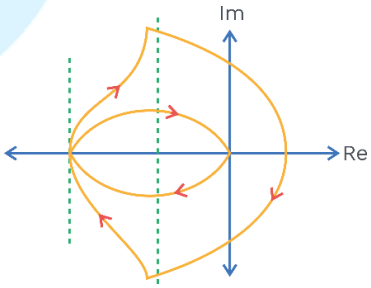


(C)



The Nyquist plot can be given as-

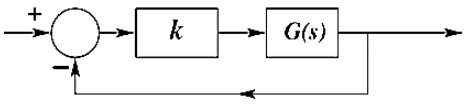
(D)



Q.2 The open loop transfer function of a unity feedback control system G(s) is unstable and has three poles on right hand side of s-plane, but the closed loop system is stable. The number of encirclements made by the Nyquist plot about (–1,0) point in the G(s) plane is _____. (Rounded off to nearest integer).

Q.3 Consider the feedback system shown in the figure. The Nyquist plot of G(s)is also shown. Which one of the

following conclusions is correct?



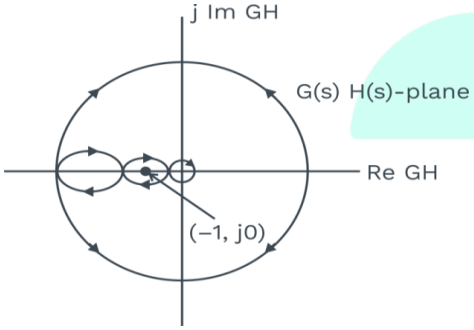
(A)G(s) is an all-pass filter

(B)G(s) is a strictly proper transfer function

(C)G(s) is a stable and minimum-phase transfer function

(D)The closed-loop system is unstable for sufficiently large and positive k

Q.4 The complete Nyquist plot of the open-loop transfer function G(s) H(s) of a feedback control system is shown in the figure.



If G(s) H(s) has one zero in the right-half of the s-plane, the number of poles that the closed-loop system will have in the right-half of the s-plane is

(A)0 (B)1

(C)4 (D)3

Q.5 The system is operating at the gain of 30 and the critical value of gain is 60. So, the gain margin (in dB) of the system is -

(A)16.01 (B)0.602

(C)3.82 (D)6.02

Q.6 For an unstable system, the phase margin and the gain margin are -

(A)Positive (B)Zero

(C)Negative (D)Not determined

Q.7 The unity feedback has an open-loop transfer function given as $\frac{20}{(s+4)^4}$. The gain margin is _____ dB.

(A)25 (B)34

(C)38 (D)42

Q.8 If the open-loop transfer function $\frac{K}{s(s+4)(s+x)}$ intersects at s = ±j20,then the phase crossover frequency (in rad/sec) is_____

(Rounded off to the nearest integer)

Q.9 The open loop transfer function of a feedback control system is-

$$G(s)\cdot H(s) = \frac{(s+1)}{s(s-3)(s-5)}$$

The gain margin of the system is _____(Up to two decimal places)

Q.10 The open loop transfer function with unity feedback is given as-

$$G(s) = \frac{K}{s(s+\alpha)^2}$$

The gain margin is 21.94 dB and the phase cross over frequency is 5 rad/sec. The value of 'K' is _____. (up to two decimal places)

Q.11 Consider a unity feedback system with open loop transfer function given as

$$G(s) = \frac{1+10s}{s^2(1+s)(1+5s)}$$

Which of the following statement(s) is/are TRUE?

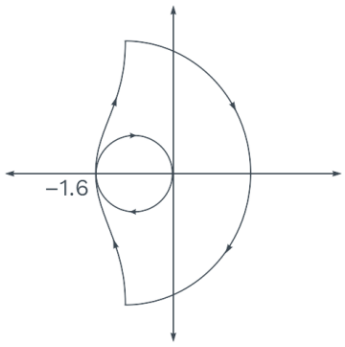
(A) The phase crossover frequency is 2.5 rad/sec

(B) The phase crossover frequency is 0.4 rad/sec

(C) The gain margin is 10.7

(D) The gain margin is 0.048

Q.12 The Nyquist plot of G(s)H(s) is given below.



IfG(s)H(s) has one right hand pole, then the corresponding closed loop system is-

(A) Stable

(B) Unstable with two righthand pole

(C) Unstable with three right hand poles

(D) Unstable with four right hand poles

Answer Key

Q.1 [D]

Q.2 [3]

Q.3 [D]

Q.4 [D]

Q.5 [D]

Q.6 [C]

Q.7 [B]

Q.8 [20, Range 19 TO 21]

Q.9 [13.33, Range 12.8 to 13.80]

Q.10 [19.99, Range 19.5 to 20.5]

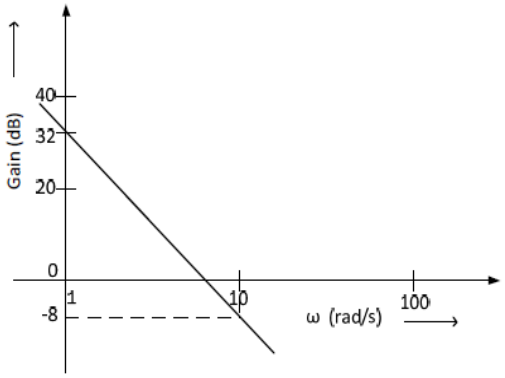
Q.11 [B, D]

Q.12 [C]

6 Bode Plot

MCQ, MSQ and NAT

Q.1 The Bode plot of a transfer function G(s) is shown in the figure below.



The gain (20 log |G(s)|) is 32 dB and 8 dB at 1 rad/s and 10 rad/s respectively. The phase is negative for all ω. Then G(s) is

(A) $\frac{39.8}{s}$ (B) $\frac{39.8}{s^2}$

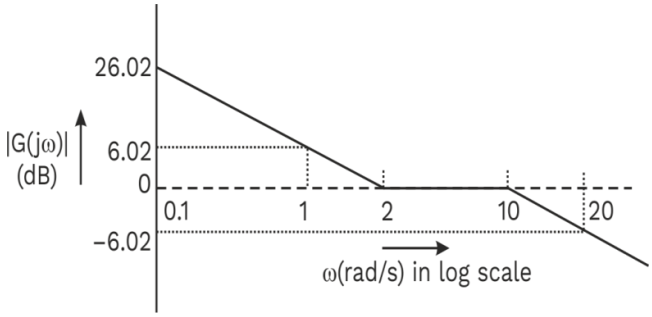
(C) $\frac{32}{s}$ (D) $\frac{32}{s^2}$

Q.2 The phase margin in degrees of

$$G(s) = \frac{10}{(s + 0.1)(s + 1)(s + 10)}$$

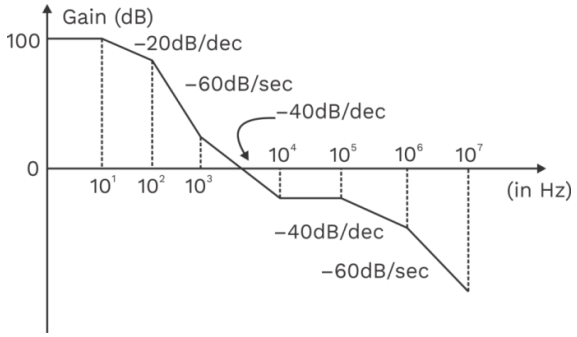
calculated using the asymptotic Bode plot is _____.

Q.3 The Bode asymptotic magnitude plot of a minimum phase system is shown in the figure.



If the system is connected in a unity negative feedback configuration, the steady state error of the closed loop system, to a unit ramp input, is _____.

Q.4 For an LTI system, the Bode plot for its gain is as illustrated in the figure shown. The number of system poles N_p and the number of system zeros N_Z in the frequency range $1Hz \leq f \leq 10^7 Hz$ is:



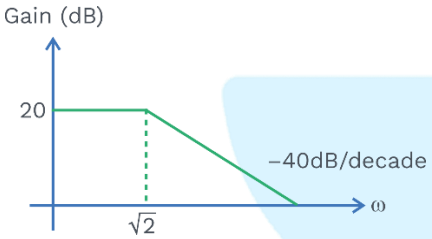
(A) $N_P = 4, N_z = 2$

(B) $N_P = 7, N_z = 4$

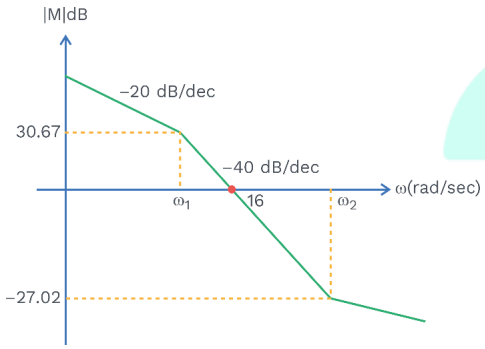
(C) $N_P = 6, N_z = 3$

(D) $N_P = 5, N_z = 2$

Q.5 The asymptotic bode plot of the transfer function $\frac{20}{s^2 + 2s + 3}$ is shown in the figure below. The magnitude of error (dB) at a frequency of $\omega = \sqrt{2}\,\text{rad/sec}$ is _____. (up to 3 decimal places)

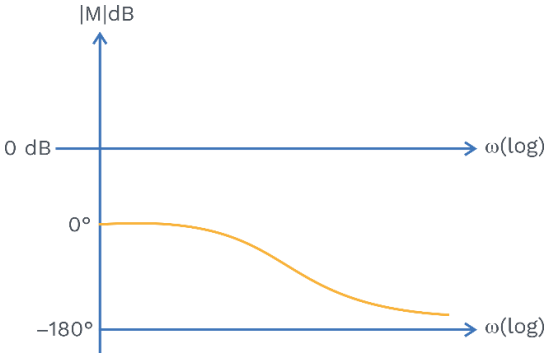


Q.6 The bode plot of a system is shown below.



The value of $\omega_1 + \omega_2$ is _____ rad/sec. (Up to 3 decimal places)

Q.7 The bode plot of a system is shown in the figure below.



The transfer function of the system is-

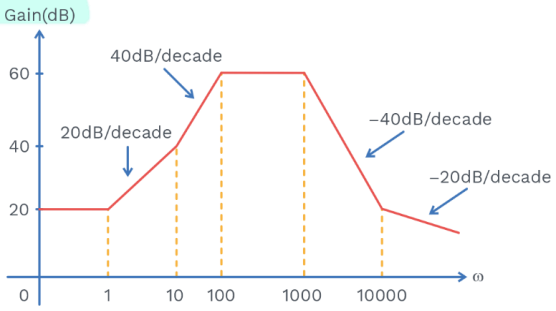
(A) $\frac{s+1}{(s-1)^2}$

(B) $\frac{1-s}{1+s}$

(C) $\frac{s}{(s+1)(s-1)}$

(D) $\frac{1}{s(s-1)^2}$

Q.8 A Bode magnitude plot shown in the figure below, the transfer function of the system is-



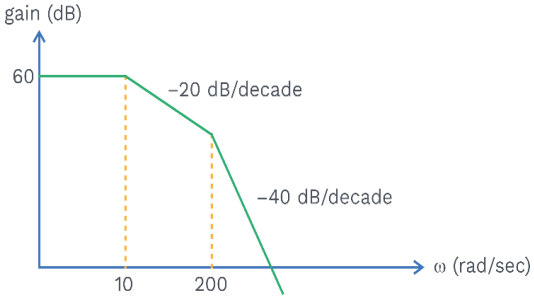
(A) $H(s) = \frac{10(s+1)(s+10)(s+10000)}{(s+100)^2(s+1000)^2}$

(B) $H(s) = \frac{10^{10}(s+1)(s+10)(s+10000)}{(s+100)^2(s+1000)^2}$

(C) $H(s) = \frac{10^{6}(s+1)(s+10)(s+1000)}{(s+100)^2(s+10000)^2}$

(D) $H(s) = \dfrac{10^6(s+1)(s+10)(s+10000)}{(s+100)^2(s+1000)^2}$

Q.9 The asymptotic bode plot is shown in the figure below.



The transfer function of the system G(s) is-

(A) $\dfrac{1000}{\left(1+\dfrac{s}{10}\right)\left(1+\dfrac{s}{200}\right)}$

(B) $\dfrac{1000}{(s+10)(s+200)}$

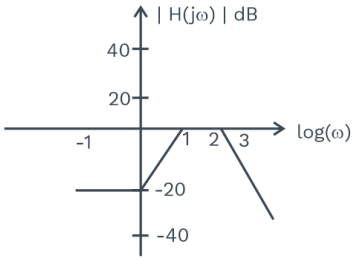
(C) $\dfrac{3}{\left(1+\dfrac{s}{10}\right)\left(1+\dfrac{s}{200}\right)}$

(D) $\dfrac{3}{(s+10)(s+250)}$

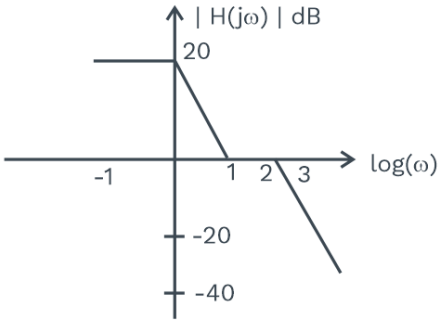
Q.10 If a transfer function consists of 1 zero and 4 poles, the final slope of the bode magnitude plot is _____ dB/decade. (Rounded off to nearest integer).

Q.11 The bode magnitude plot $H(j\omega) = \dfrac{10^4(1+j\omega)}{(10+j\omega)(100+j\omega)}$ is

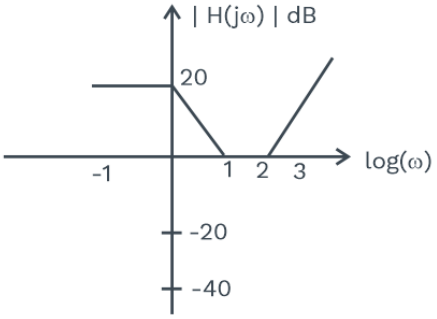
(A)



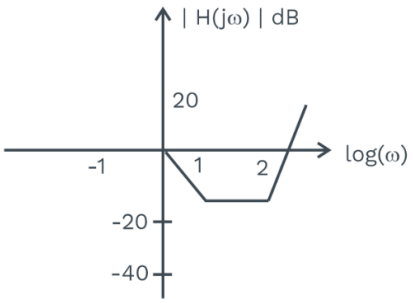
(B)



(C)

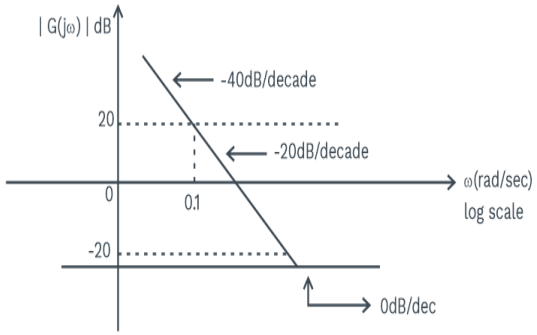


(D)



Q.12 The Asymptotic Bode Magnitude plot of a minimum phase transfer function

is shown below. The difference between (P – Z) poles and zeroes is _____



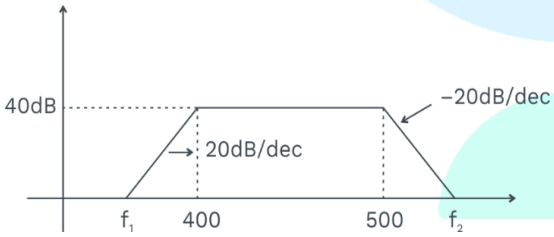
(A) 0

(B) 1

(C) 2

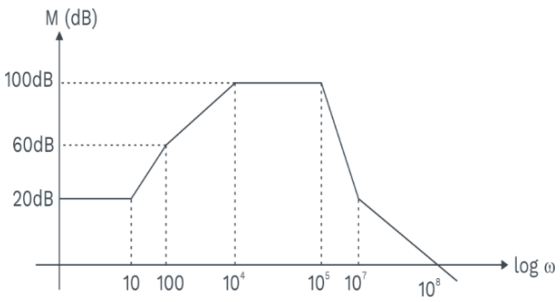
(D) 3

Q.13 For the bode plot given, assume all poles and zeros are real valued.



Then the value of $f_2 - f_1$ is _____ kHz.(Upto 2 decimal places)

Q.14 The transfer function for minimum phase system whose the bode magnitude plot given below is -



(A)

$$\frac{\left(1+\frac{s}{10}\right)^2\left(1+\frac{s}{10^7}\right)}{\left(1+\frac{s}{100}\right)\left(1+\frac{s}{10^4}\right)\left(1+\frac{s}{10^5}\right)^2}$$

(B)

$$\frac{10\left(1+\frac{s}{10}\right)^2\left(1+\frac{s}{10^5}\right)}{\left(1+\frac{s}{100}\right)\left(1+\frac{s}{10^4}\right)\left(1+\frac{s}{10^7}\right)^2}$$

(C)

$$\frac{10\left(1+\frac{s}{10}\right)^2\left(1+\frac{s}{10^7}\right)}{\left(1+\frac{s}{100}\right)\left(1+\frac{s}{10^4}\right)\left(1+\frac{s}{10^5}\right)^2}$$

(D)

$$\frac{10\left(1+\frac{s}{10}\right)^2\left(1+\frac{s}{10^5}\right)}{\left(1+\frac{s}{100}\right)\left(1+\frac{s}{10^4}\right)\left(1+\frac{s}{10^7}\right)^2}$$

Answer Key

Q.1 [B]

Q.2 [45, Range 42 to 48]

Q.3 [0.5, Range 0.45 to 0.55]

Q.4 [C]

Q.5 [3.010, Range 3 to 3.020]

Q.6 [78.528, Range 77 to 79]

Q.7 [B]

Q.8 [D]

Q.9 [A]

Q.10 [-60, Range -61 to -59]

Q.11 [A]

Q.12 [C]

Q.13 [49.99, Range 49.5 to 50.5]

Q.14 [C]

7 State Space Analysis

MCQ, MSQ and NAT

Q.1 The state variable description of an LTI system is given by

$$\begin{pmatrix}\dot{x}_1\\ \dot{x}_2\\ \dot{x}_3\end{pmatrix}=\begin{pmatrix}0 & a_1 & 0\\ 0 & 0 & a_2\\ a_3 & 0 & 0\end{pmatrix}\begin{pmatrix}x_1\\ x_2\\ x_3\end{pmatrix}+\begin{pmatrix}0\\ 0\\ 1\end{pmatrix}u$$

$$y=\begin{pmatrix}1 & 0 & 0\end{pmatrix}\begin{pmatrix}x_1\\ x_2\\ x_3\end{pmatrix}$$

where y is the output and u are the input? The system is controllable for

(A) $a_1 \neq 0, a_2 = 0, a_3 \neq 0$

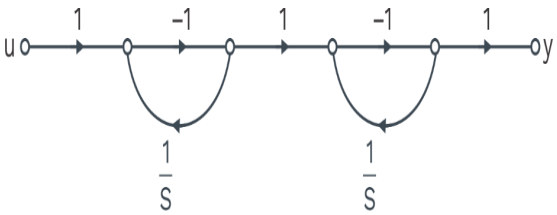
(B) $a_1 = 0, a_2 \neq 0, a_3 \neq 0$

(C) $a_1 = 0, a_2 \neq 0, a_3 = 0$

(D) $a_1 \neq 0, a_2 \neq 0, a_3 = 0$

Q.2 The state diagram of a system is shown below. A system is described by the state-variable equations

$\dot{X} = AX + Bu;\ y = CX + Du$



The state-variable equations of the system shown in the figure above are

(A) $\dot{X} = \begin{bmatrix}-1 & 0\\ 1 & -1\end{bmatrix}X + \begin{bmatrix}-1\\ 1\end{bmatrix}u$

$y = [1\ \ -1]\,X + u$

(B) $\dot{X} = \begin{bmatrix}-1 & 0\\ -1 & -1\end{bmatrix}X + \begin{bmatrix}-1\\ 1\end{bmatrix}u$

$y = [-1\ \ -1]\,X + u$

(C) $\dot{X} = \begin{bmatrix}-1 & 0\\ -1 & -1\end{bmatrix}X + \begin{bmatrix}-1\\ 1\end{bmatrix}u$

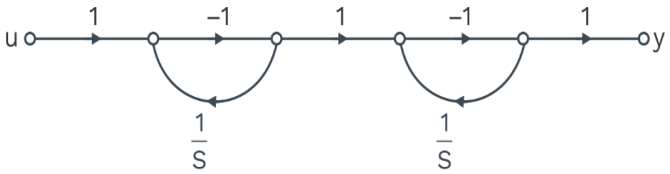
$y = [-1\ \ -1]\,X - u$

(D) $\dot{X} = \begin{bmatrix}-1 & -1\\ 0 & -1\end{bmatrix}X + \begin{bmatrix}-1\\ 1\end{bmatrix}u$

$y = [1\ \ -1]\,X - u$

Q. 3 The state diagram of a system is shown below. A system is described by the state-variable equations

$\dot{X} = AX + Bu;\ y = CX + Du$



The state transition matrix e^{At} of the system shown in the figure above is

(A) $\begin{bmatrix} e^{-t} & 0 \\ te^{-t} & e^{-t} \end{bmatrix}$

(B) $\begin{bmatrix} e^{-t} & 0 \\ -te^{-t} & e^{-t} \end{bmatrix}$

(C) $\begin{bmatrix} e^{-t} & 0 \\ e^{-t} & e^{-t} \end{bmatrix}$

(D) $\begin{bmatrix} e^{-t} & -te^{-t} \\ 0 & e^{-t} \end{bmatrix}$

Q.4 Consider the state space model of a system, as given below

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \dot{x}_3 \end{bmatrix} = \begin{bmatrix} -1 & 1 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & -2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} + \begin{bmatrix} 0 \\ 4 \\ 0 \end{bmatrix} u; y = [1\ 1\ 1] \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$$

.

The system is

(A)controllable and observable

(B)uncontrollable and observable

(C)uncontrollable and unobservable

(D)controllable and unobservable

Q.5

An unforced linear time invariant (LTI) system is represented by

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} = \begin{bmatrix} -1 & 0 \\ 0 & -2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

If the initial conditions are $x_1(0) = 1$ and $x_2(0) = -1$, the solution of the state equation is

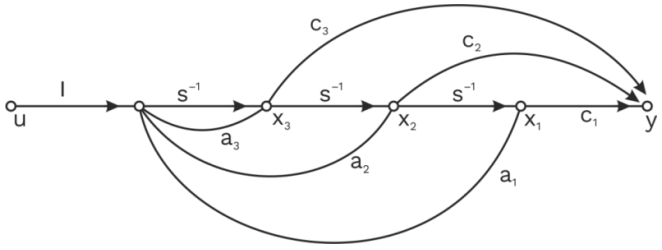
(A) $x_1(t) = -1,\ x_2(t) = 2$

(B) $x_1(t) = -e^{-t},\ x_2(t) = 2e^{-t}$

(C) $x_1(t) = e^{-t},\ x_2(t) = -e^{-2t}$

(D) $x_1(t) = -e^{-t},\ x_2(t) = -2e^{-t}$

Q.6 Consider the state space system expressed by the signal flow diagram shown in the figure.



The corresponding system is

(A)always controllable

(B)always observable

(C)always stable

(D)always unstable

Q.7 The state equation of a second-order linear system is given by

$$\dot{x}(t) = Ax(t), x(0) = x_0$$

For $\mathbf{x}_0 = \begin{bmatrix} 1 \\ -1 \end{bmatrix}, \mathbf{x}(t) = \begin{bmatrix} e^{-t} \\ -e^{-t} \end{bmatrix}$ and for $\mathbf{x}_0 = \begin{bmatrix} \&0 \\ \&1 \end{bmatrix}, \mathbf{x}(t) = \begin{bmatrix} e^{-t} - e^{-2t} \\ -e^{-t} + 2e^{-2t} \end{bmatrix}$

When $x_0 = \begin{bmatrix} \&3 \\ \&5 \end{bmatrix}$, x(t) is.

(A) $\begin{bmatrix} -8e^{-t} + 11e^{-2t} \\ 8e^{-t} - 22e^{-2t} \end{bmatrix}$

(B) $\begin{bmatrix} 11e^{-t} - 8e^{-2t} \\ -11e^{-t} + 16e^{-2t} \end{bmatrix}$

(C) $\begin{bmatrix} 3e^{-t} - 5e^{-2t} \\ -3e^{-t} + 10e^{-2t} \end{bmatrix}$

(D) $\begin{bmatrix} 5e^{-t} - 3e^{-2t} \\ -5e^{-t} + 6e^{-2t} \end{bmatrix}$

Q.8 Let the state-space representation of an LTI system be $\dot{x}(t) = Ax(t) + Bu(t)$,

$y(t) = Cx(t) + du(t)$ where A, B, C are matrices, d is a scalar, u(t) is the input to the system, and y(t) is its output. Let $B = [0\ 0\ 1]^T$ and d = 0. Which one of the following options for A and C will ensure that the transfer function of this LTI system is $H(s) = \frac{1}{s^3 + 3s^2 + 2s + 1}$?

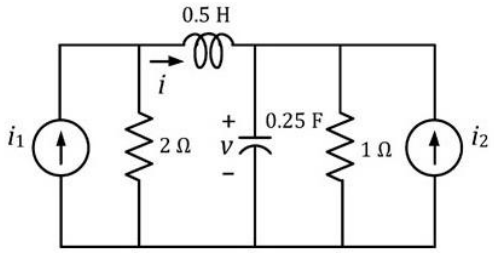
(A) $A = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -1 & -2 & -3 \end{bmatrix}$ and $C = [1\ 0\ 0]$

(B) $A = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -1 & -2 & -3 \end{bmatrix}$ and $C = [0\ 0\ 1]$

(C) $A = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -3 & -2 & -1 \end{bmatrix}$ and $C = [0\ 0\ 1]$

(D) $A = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -3 & -2 & -1 \end{bmatrix}$ and $C = [1\ 0\ 0]$

Q.9 For the given circuit, which one of the following is the correct state equation?



(A)

$$\frac{d}{dt}\begin{bmatrix} v \\ i \end{bmatrix} = \begin{bmatrix} -4 & 4 \\ -2 & -4 \end{bmatrix}\begin{bmatrix} v \\ i \end{bmatrix} + \begin{bmatrix} 0 & 4 \\ 4 & 0 \end{bmatrix}\begin{bmatrix} i_1 \\ i_2 \end{bmatrix}$$

(B)

$$\frac{d}{dt}\begin{bmatrix} v \\ i \end{bmatrix} = \begin{bmatrix} -4 & -4 \\ -2 & 4 \end{bmatrix}\begin{bmatrix} v \\ i \end{bmatrix} + \begin{bmatrix} 4 & 4 \\ 4 & 0 \end{bmatrix}\begin{bmatrix} i_1 \\ i_2 \end{bmatrix}$$

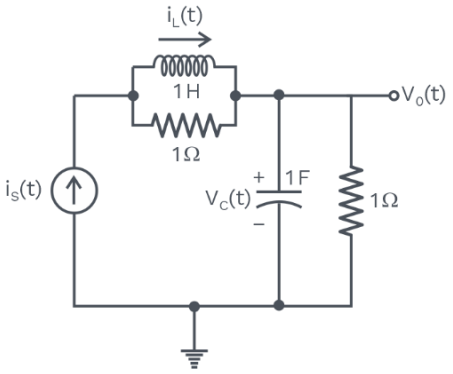
(C)

$$\frac{d}{dt}\begin{bmatrix} v \\ i \end{bmatrix} = \begin{bmatrix} 4 & -4 \\ -2 & -4 \end{bmatrix}\begin{bmatrix} v \\ i \end{bmatrix} + \begin{bmatrix} 0 & 4 \\ 4 & 4 \end{bmatrix}\begin{bmatrix} i_1 \\ i_2 \end{bmatrix}$$

(D)

$$\frac{d}{dt}\begin{bmatrix} v \\ i \end{bmatrix} = \begin{bmatrix} -4 & -4 \\ -2 & -4 \end{bmatrix}\begin{bmatrix} v \\ i \end{bmatrix} + \begin{bmatrix} 4 & 0 \\ 0 & 4 \end{bmatrix}\begin{bmatrix} i_1 \\ i_2 \end{bmatrix}$$

Q.10 The electrical system shown in the figure converts input source current $i_s(t)$ to output voltage $v_0(t)$.



Current $i_L(t)$ in the inductor and voltage $v_C(t)$ across the capacitor are taken as the state variables, both assumed to be initially equal to zero, i.e., $i_L(0) = 0$ and $v_C(0) = 0$. The system is

(A)completely state controllable as well as completely observable

(B)completely state controllable but not observable

(C)completely observable but not state controllable

(D)neither state controllable nor observable

Q.11 The state matrix in a state space representation is given by $A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 3 \end{bmatrix}$. The closed loop poles of a system having state matrix $[A]^{-1}$ will be

(A) 1, 2, 3

(B) -1, -2, -3

(C) $1, \frac{1}{2}, \frac{1}{3}$

(D) 1, 4, 9

Q.12 The state model of a system is given as

$$\dot{X} = \begin{bmatrix} 0 & 1 \\ -2 & -2 \end{bmatrix} X + \begin{bmatrix} 1 \\ 2 \end{bmatrix} u$$

$Y = [1 \quad -1]x$

The above system is

(A) A non-minimum phase system

(B) A minimum phase system

(C) An all-pass system

(D) None of the above

Q.13 A linear time invariant system of second order has state space equations -

$$\dot{x}_1 = -4x_1 - x_2 + 3u$$

$$\dot{x}_2 = -4x_2 - u$$

$y = 3x_1 - 2x_2$

Then the output response of the system is -

(A)Critically damped

(B)Underdamped

(C)Undamped

(D)Overdamped

Q.14 The state space representation of a system is given by-

$$\dot{x} = \begin{bmatrix} -3 & 0 \\ 1 & -2 \end{bmatrix} + \begin{bmatrix} 0 \\ 1 \end{bmatrix} u$$

$y = [1 \quad 1]x$

the system is-

(A) Controllable and observable

(B) Uncontrollable and Unobservable

(C) Uncontrollable and observable

(D) Controllable and unobservable

Q.15 The state space representation of a system is given as-

$$\dot{x} = \begin{bmatrix} 0 & 1 \\ 2 & -1 \end{bmatrix} x + \begin{bmatrix} 1 \\ 0 \end{bmatrix} u$$

$y = [\,0 \quad 1]x$

then the transfer function of the system $\frac{Y(s)}{U(s)}$ will be-

(A) $\frac{2(s+1)}{s^2+s+2}$

(B) $\frac{s+1}{s^2+s-2}$

(C) $\frac{2(s+1)}{s^2+s-2}$

(D) $\frac{2}{s^2+s-2}$

Q.16 Determine the characteristic equation of the given matrix,

$$A = \begin{bmatrix} 0 & -2 & 1 \\ 2 & -1 & 0 \\ 3 & 2 & 1 \end{bmatrix}$$

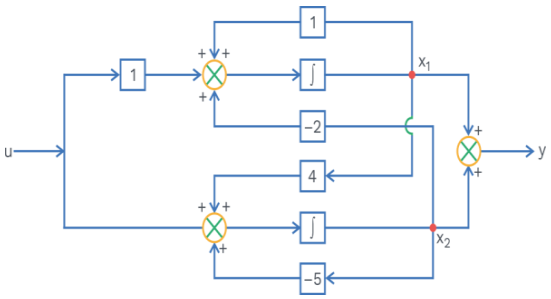
(A) $s^3 - 9 = 0$

(B) $s^3 - s + 9 = 0$

(C) $s^3 - s - 9 = 0$

(D) $s^3 - 7s - 9 = 0$

Q.17 For a control system the state block diagram is given below as-



The system represented above is said to be-

(A) Controllable and observable

(B) Observable but not controllable

(C) Controllable but not observable

(D) Neither controllable nor observable

Q.18 A system is described by the state model as shown below.

$$\dot{x} = \begin{bmatrix} -1 & 1 & 1 \\ 1 & 0 & -1 \\ -1 & 1 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} + \begin{bmatrix} 0 \\ -1 \\ 0 \end{bmatrix} [u]$$

$y = [\,0 \quad -1 \quad 1]x$

Find the number of closed-loop poles having positive real part.

Q.19 A control system is represented by the state model as-

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} = \begin{bmatrix} -2 & 0 \\ 0 & -8 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} 0 \\ 2 \end{bmatrix} u$$

The response of the system for the unit step input is-

(A) Overdamped

(B) Underdamped

(C) Critically damped

(D) Undamped

Q.20 Consider a system G(s) represented by the state model given below.

$$A = \begin{bmatrix} 0 & 3 \\ -1 & -2 \end{bmatrix}, \quad B = \begin{bmatrix} 0 \\ 1 \end{bmatrix} \text{ and}$$

$C = [\,2 \quad 2]$

If unity feedback is provided to G(s), then which of the given

statement(s) is/are true for the system?

(A)The steady-state error for the unit ramp is infinite.

(B)The steady-state error for the unit ramp is zero.

(C)The steady-state error for the unit step is 3.

(D)The steady-state error for the unit step is 0.33.

Q.21 The forward path transfer function $G(s)=\begin{bmatrix}\frac{1}{s+1} & -1\\ 1 & \frac{1}{s+2}\end{bmatrix}$ is and the transfer function of the feedback path is defined as $H(s)=\begin{bmatrix}1 & 0\\ 0 & 1\end{bmatrix}$. Then which of the following statement(s) is/are true?

(A)The transfer function of the system is $T(s)=\frac{s^2+3s+3}{s^2+5s+6}$

(B)The transfer function of the system is $T(s)=\frac{s^2+s-3}{s^2+5s+6}$.

(C)The damping ratio of the system is 0.416.

(D)The damping ratio of the system is 1.02.

Q.22 The eigenvalues of the matrix $A=\begin{bmatrix}0 & 1 & -1\\ -6 & -11 & 6\\ -6 & -11 & 5\end{bmatrix}$ are:

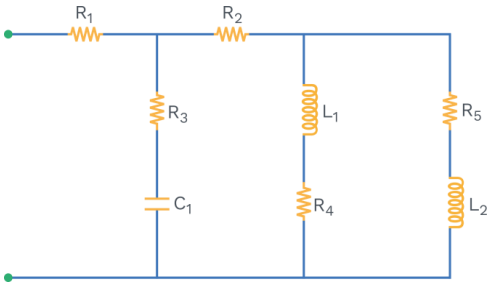
(A)–1, –2 and –3

(B)6, 11 and –5

(C)–16, 5 and –11

(D)None of the above

Q.23 A network is shown in the figure below:



The minimum number of states required to describe the network is _____.

Q.24 A linear system, $\dot{x}=\begin{bmatrix}-2 & 0\\ 0 & 1\end{bmatrix}x$ is having an initial condition as $x(0)=\begin{bmatrix}1\\ 2\end{bmatrix}$. The system x(t) is-

(A) $\begin{bmatrix}e^{-2t} & 0\\ 0 & e^{t}\end{bmatrix}$

(B) $\begin{bmatrix}e^{-t} & 0\\ 0 & e^{2t}\end{bmatrix}\begin{bmatrix}2\\ 1\end{bmatrix}$

(C) $\begin{bmatrix}e^{2t} & 0\\ 0 & e^{-t}\end{bmatrix}$

(D) $\begin{bmatrix}e^{-2t} & 0\\ 0 & e^{t}\end{bmatrix}\begin{bmatrix}1\\ 2\end{bmatrix}$

Q.25 The representation of a system in state space is $\dot{X}=Ax+Bu$, where

$A = \begin{bmatrix} 2 & 4 \\ a & 8 \end{bmatrix}$ and $B = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$.

The value of 'a' for which the system is uncontrollable is _____ .(Rounded off to the nearest integer)

Q.26 The state transition matrix is $\dot{x}(t) = \begin{bmatrix} 0 & 3 \\ -1 & -4 \end{bmatrix} x(t)$. The inverse of the state transition matrix is-

(A) $\begin{bmatrix} -e^{3t} + 3e^{t} & -3e^{3t} + 3e^{t} \\ e^{3t} - e^{t} & 3e^{-3t} + e^{t} \end{bmatrix}$

(B) $\begin{bmatrix} -\frac{1}{2}e^{3t} + \frac{3}{2}e^{t} & -\frac{3}{2}e^{3t} + \frac{3}{2}e^{t} \\ \frac{1}{2}e^{3t} - \frac{1}{2}e^{t} & \frac{3}{2}e^{3t} - \frac{1}{2}e^{t} \end{bmatrix}$

(C) $\begin{bmatrix} -\frac{1}{2}e^{-3t} + \frac{3}{2}e^{-t} & -\frac{3}{2}e^{-3t} + \frac{3}{2}e^{-t} \\ \frac{1}{2}e^{-3t} - \frac{1}{2}e^{-t} & \frac{3}{2}e^{-3t} + \frac{1}{2}e^{-t} \end{bmatrix}$

(D) $\begin{bmatrix} -\frac{3}{2}e^{3t} + \frac{1}{2}e^{t} & \frac{3}{2}e^{3t} - \frac{3}{2}e^{t} \\ \frac{1}{2}e^{-3t} - \frac{1}{2}e^{-t} & \frac{3}{2}e^{-3t} + \frac{1}{2}e^{-t} \end{bmatrix}$

Q.27 A difference equation is used to represent a discrete system as-

$$X(k+1) = \begin{bmatrix} \alpha - 1 & \alpha + 5 \\ \alpha & \alpha \end{bmatrix} X(k)$$

The initial conditions $X_1(0) = X_2(0) = 0$. If $\alpha = 2$,then the position of pole that makes the system unstable is _____ . (Up to three decimal places)

Q.28 The transfer function of a system is given by $T(s) = \frac{s + 30}{s^2 + 11s + 30}$. The state-space representation of the transfer function is $\dot{x} = Ax + Bu$ where $A = \begin{bmatrix} 0 & 1 \\ -30 & -11 \end{bmatrix}$ and $B = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$.

The sum of the elements of the matrix C is-

(A) 0 (B) 1

(C) 31 (D) 42

Q.29 The matrices of a state space model are given by $A = \begin{bmatrix} 0 & 4 \\ -2 & -6 \end{bmatrix}$, $B = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$ and

$C = [1 \quad 0]$.

The location of poles of the system is-

(A) –4, –2 (B) 0, –4

(C) –2, –6 (D) 4, 2

Q.30 The state-space model of the two systems is given below.

SYSTEM 1:

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ -5 & -6 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} 0 \\ 1 \end{bmatrix} u_1(t)$$

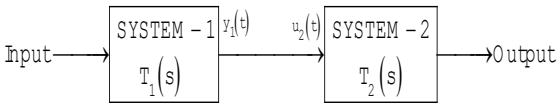
$$y_1 = \begin{bmatrix} 1 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

SYSTEM 2:

$$\begin{bmatrix} \dot{x}_3 \\ \dot{x}_4 \end{bmatrix} = \begin{bmatrix} -4 & 1 \\ 0 & -3 \end{bmatrix} \begin{bmatrix} x_3 \\ x_4 \end{bmatrix} + \begin{bmatrix} 0 \\ 2 \end{bmatrix} u_2(t)$$

$$y_2 = \begin{bmatrix} 1 & 2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

The two systems are cascaded as shown below.



The transfer function of the resultant system is-

(A) $T(s) = \dfrac{2}{s^2 + 7s + 11}$

(B) $T(s) = \dfrac{4s + 18}{s^3 + 12s^2 + 47s + 60}$

(C) $T(s) = \dfrac{4(s+1)}{s^4 + 13s^3 + 59s^2 + 107s + 60}$

(D) $T(s) = \dfrac{2(2s+9)(s+5)}{s^4 + 13s^3 + 59s^2 + 107s + 60}$

Q.31 A linear time-invariant system has the state and the output equations given as-

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} = \begin{bmatrix} 2 & -1 \\ 0 & 3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} 0 \\ 1 \end{bmatrix} u$$

$$y = \begin{bmatrix} 1 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

If the initial conditions are given as $x_1(0) = 3$, $x_2(0) = -2$ and $u(0) = 0$.

The value of $\dfrac{dy(t)}{dt}$ at $t = 0$ is _____ . (Rounded off to the nearest integer)

Q.32 The state-space representation of a system is given by-

$$\dot{x} = \begin{bmatrix} -5 & 0 \\ 1 & -4 \end{bmatrix} + \begin{bmatrix} 0 \\ 1 \end{bmatrix} u$$

$y = [1 \quad -1]x$

The system is-

(A) Controllable

(B) Observable

(C) Uncontrollable

(D) Unobservable

Q.33 Determine the characteristic equation of the given matrix.

$$A = \begin{bmatrix} 0 & -3 & 1 \\ 3 & -2 & 0 \\ 5 & 1 & 2 \end{bmatrix}$$

(A) $s^3 - 31 = 0$

(B) $s^3 - 18s + 31 = 0$

(C) $2s^3 - 6s - 9 = 0$

(D) $s^3 - 12s - 18 = 0$

Q.34 Find the state transition matrix for the given matrix, where X and Y are the real numbers.

$$A = \begin{bmatrix} X & -Y \\ Y & X \end{bmatrix}$$

(A)

$$\phi(t) = \begin{bmatrix} e^{Xt}\cos Yt & -e^{Xt}\sin Yt \\ e^{Xt}\sin Yt & e^{Xt}\cos Yt \end{bmatrix}$$

(B)

$$\phi(t) = \begin{bmatrix} e^{-Xt}\sin Yt & e^{Xt}\cos Yt \\ -e^{Xt}\cos Yt & e^{-Xt}\sin Yt \end{bmatrix}$$

(C)

$$\phi(t) = \begin{bmatrix} e^{-\sigma t}\cos\omega t & e^{\sigma t}\sin\omega t \\ -e^{\sigma t}\sin\omega t & e^{-\sigma t}\cos\omega t \end{bmatrix}$$

(D)

$$\phi(t) = \begin{bmatrix} e^{\sigma t}\sin\omega t & -e^{\sigma t}\cos\omega t \\ e^{\sigma t}\cos\omega t & e^{\sigma t}\sin\omega t \end{bmatrix}$$

Q.35 A system is represented by $\dot{X} = AX + BU$, where $A = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -5 & -K & -11 \end{bmatrix}$. The range of K for stability is-

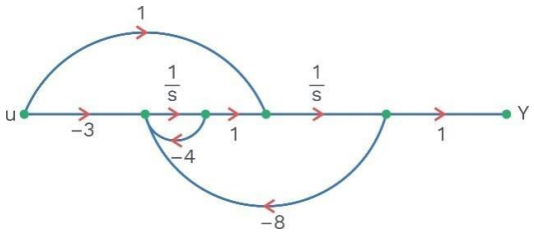
(A) $0 < K < \frac{5}{11}$

(B) $K > \frac{5}{11}$

(C) $K > \frac{11}{5}$

(D) $K < \frac{5}{11}$

Q.36 The signal flow diagram is shown in the figure below.



The determinant for observability is _____. (Rounded off to nearest integer).

Answer Key

Q.1 [D]

Q.2 [A]

Q.3 [A]

Q.4 [B]

Q.5 [C]

Q.6 [A]

Q.7 [B]

Q.8 [A]

Q.9 [A]

Q.10 [D]

Q.11 [C]

Q.12 [A]

Q.13 [A]

Q.14 [B]

Q.15 [D]

Q.16 [A]

Q.17 [B]

Q.18 [1]

Q.19 [A]

Q.20 [A, D]

Q.21 [A, D]

Q.22 [A]

Q.23 [3]

Q.24 [D]

Q.25 [-2, Range -2.1 to -1.9]

Q.26 [B]

Q.27 [5.274, Range 5.05 to 5.39]

Q.28 [C]

Q.29 [A]

Q.30 [B]

Q.31 [2, Range 1.9 To 2.1]

Q.32 [B, C]

Q.33 [A]

Q.34 [A]

Q.35 [B]

Q.36 [1]

8 Controllers and Compensators

MCQ, MSQ and NAT

Q.1 $G_c(s)$ is a lead compensator if

(A) $a = 1, b = 2$

(B) $a = 3, b = 2$

(C) $a = -3, b = -1$

(D) $a = 3, b = 1$

Q.2 The phase of the above lead compensator is maximum at

(A) $\sqrt{2}\ rad/s$

(B) $\sqrt{3}\ rad/s$

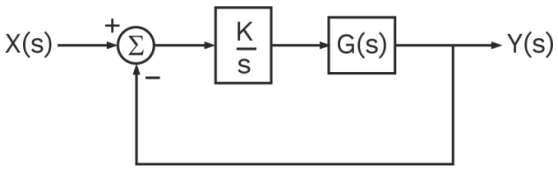
(C) $\sqrt{6}\ rad/s$

(D) $1/\sqrt{3}\ rad/s$

Q.3 Consider a unity feedback system, as in the figure shown, with an integral compensator $\frac{k}{s}$ and open-loop transfer function

$$G(s) = \frac{1}{s^2 + 3s + 2}$$

Where K > 0. The positive value of K for which there are exactly two poles of the unity feedback system on the j□□axis is equal to _____ (rounded off to two decimal places).



Q.4 Consider the following statements:

I. The steady state accuracy is improved by lag networks.

II. Phase lead compensators can be used to increase the bandwidth.

III. By using phase lag compensators. The speed of the response is reduced.

IV. Phase lead compensation can be used to reduce settling time.

Which of the above statements are correct?

(A) I, II and III

(B) III and IV

(C) I, II and IV

(D) All the statements

Q.5 The transfer function of a network is –

$$G(s) = \frac{1 + \frac{s}{16}}{1 + \frac{s}{49}}$$

Then the frequency (in rad/sec) at which the network will provide maximum phase lead is -

(A)28 (B)3.06

(C)1.75 (D)16.81

Q.6 Effects of integral error compensation on the various parameters in a control system is -

(A) It does not have any effect on steady state response.

(B) The steady state error is reduced

(C) It decreases the type of the system

(D) It makes the system more stable

Q.7 The maximum phase shift of the compensator $G_c(s) = \frac{5(3+0.9s)}{4+0.4s}$ is-

(A) 30° (B) 60°

(C) 45° (D) 90°

Q.8 Consider the following statement about the compensates:

1. Pole is dominant in lag compensator.
2. Pole is dominant in lead compensator.
3. Zero is dominant in lag compensator.
4. Zero is dominant in lead compensator.

Which of the above statement(s) is/are correct?

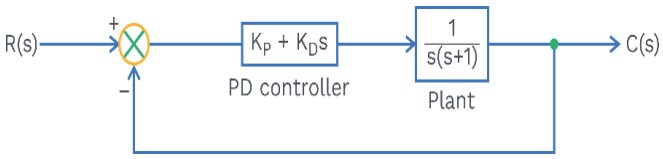
(A) 1 and 4

(B) 2 and 3

(C) 1 and 3

(D) 2 and 4

Q.9 The block diagram of a control system is shown below:



The damping ratio of the closed-loop system is 0.8 and the damped natural frequency is 3rad/sec. The parameters of PD controller are K_P and K_D. The value of $\frac{K_P}{K_D}$ is _____ (Up to three decimal places)

Q.10 The open-loop transfer function of a unity feedback control system is given as $G(s) = \frac{1}{s(s+2)}$. If the first-order compensator is used in the system, then the poles of the closed-loop system are –2 ± 4j and –8. The transfer function of the compensator is-

(A) $5\left(\frac{s+10}{s+5}\right)$

(B) $32\left(\frac{s+5}{s+10}\right)$

(C) $\frac{(s+9)(s+5)}{5(s+10)}$

(D) $\frac{1}{(s+50)(s+10)}$

Q.11 The transfer function of the two compensators are given as-

$G_1 = \frac{10(s+2)}{(s+5)}$, $G_2 = \frac{(s+5)}{10(s+1)}$

Which one of the following statements is correct?

(A) Both G_1 and G_2 are lag compensator.

(B) Both G_1 and G_2 are lead compensator.

(C) G_1 is lead compensator and G_2 is a lag compensator.

(D) G_1 is lag compensator and G_2 is lead compensator.

Answer Key

Q.1 [A]

Q.2 [A]

Q.3 [6, Range 5.99 to 6.01]

Q.4 [D]

Q.5 [A]

Q.6 [B]

Q.7 [A]

Q.8 [A]

Q.9 [3.571, Range 3.45 to 3.62]

Q.10 [B]

Q.11 [C]